

COVARIANT STRETCHING–DIFFUSION RIGIDITY AND GLOBAL REGULARITY FOR THE THREE-DIMENSIONAL NAVIER–STOKES EQUATIONS

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ABSTRACT. We formulate global regularity as a covariant no-loss principle for the three-dimensional incompressible Navier–Stokes equations. If X is the volume-preserving material flow, $F = \nabla X$, and $g = F^T F$, then vortex stretching and viscous diffusion are generated by the dual tensors F and g^{-1} . Amplification in one material direction therefore forces compression, and hence an enhanced diffusive direction, in another. We prove the exact pullback identities expressing this stretching–diffusion duality and isolate the equality alternative: a target-retaining orbit must either expend covariant dissipation or approach a stretching-depleted material geometry.

The regularity argument is organized around one no-loss cascade theorem. A terminal singularity produces infinitely many compact normalized windows. The no-loss theorem assigns a positive amount of one finite covariant expenditure measure to every such window, unless its material limit is oscillation-removable. Compactness and equality rigidity then exclude the singular orbit in a short argument. In the phase-I manuscript the companion trilogy supplies this theorem: its branch closures and moving-cutoff carrier combine to produce the required measure, while zero carrier production forces a constant local velocity. The independent version is reduced to a signed scale-space flux identity and classification of lossless material geometries, recorded as explicit appendix contracts. Backward adjoint densities and the covariant heat scale provide a Perelman-type diagnostic for the equality geometry; they are not used to estimate pressure term by term.

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1. INTRODUCTION

Consider the incompressible Navier–Stokes equations on $\mathbb{R}^3$,

$$(1.1) \quad \partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \operatorname{div} u = 0, \quad \nu > 0.$$

Three geometric observations organize the argument. Arnold’s description of Euler flow treats incompressible transport as conservative motion on the group of volume-preserving diffeomorphisms [1, 6]. Onsager’s variational description of irreversible evolution identifies a dissipative current with a gradient in a mobility metric [9]. For Navier–Stokes these statements fit together exactly:

$$(1.2) \quad \underbrace{\partial_t u + \mathbb{P}((u \cdot \nabla)u)}_{\text{Euler-covariant derivative}} = - \underbrace{\nu(-\mathbb{P}\Delta)}_{\text{Stokes mobility}} D\mathcal{E}(u), \quad \mathcal{E}(u) = \frac{1}{2}\|u\|_2^2.$$

The conservative term is orthogonal to $D\mathcal{E}(u) = u$, while the squared metric norm of the right-hand side is $\nu\|\nabla u\|_2^2$. Thus the energy inequality is the energy–dissipation identity associated with (1.2).

The third observation is specific to incompressible vorticity. In material coordinates the deformation tensor $F = \nabla_a X$ transports vorticity, while viscosity is pulled back by the inverse Cauchy–Green metric $\mathbf{g}^{-1} = (F^T F)^{-1}$. Since $\det F = 1$, stretching and enhanced diffusion are dual aspects of the same volume-preserving deformation. This is the structural mechanism developed in Section 4.

The physical Stokes action alone does not exclude concentration because its cost on a normalized event at scale r is weighted by r . The proof therefore organizes the cross-scale transfer as one covariant expenditure rather than estimating every Eulerian branch independently. Its steps are:

1. A terminal loss of regularity generates a coherent forward orbit in repaired scale–translation coordinates. The orbit is normalized modulo translation, dilation, and pressure gauge. Its scale–center slice and observables remove Galilean modes by fixed-cutoff centering.
2. The covariant no-loss theorem assigns one finite expenditure measure to the orbit. A zero-cost material limit is stretching-depleted and belongs to the oscillation-removable class.
3. Compactness upgrades this equality classification to a positive cost on every window retaining scale-critical velocity oscillation.
4. Infinitely many disjoint retained windows would have infinite total cost, contradicting finiteness of the covariant expenditure.

This separation follows directly from the localized Navier–Stokes energy identity. Its convection, pressure, cutoff-motion, and chart terms depend linearly on the localization test. For a compatible parent–child partition they are therefore internal signed currents and cancel before estimates are taken. Viscous expenditure and the suitable-energy defect have a fixed sign. Section 10 develops this accounting principle entirely within the Navier–Stokes equation and isolates the remaining material rigidity problem.

In the phase-I proof, the corrected moving-cutoff energy realizes the covariant expenditure after the companion trilogy has routed all nonsummable fluxes. This construction proves the

no-loss theorem without adding a new regularity assumption. The independent proof seeks the same measure directly: convection, pressure, and chart motion are retained as signed internal currents on a nested scale-space tree, and only exterior input and viscous loss survive summation. The remaining equality case is a material rigidity problem rather than a list of pressure and modulation estimates.

Theorem 1.1 (Covariant no-loss criterion for regularity). *Assume that the canonical critical-orbit bridge [Bridge hypothesis 9.2](#) and the covariant no-loss cascade bridge [Bridge hypothesis 10.3](#) hold for the class of global finite-energy suitable weak solutions of (1.1). Then every such solution satisfies*

$$\Sigma_u(T) = \emptyset \quad \text{for every finite } T > 0.$$

Consequently every divergence-free datum $u_0 \in H^m(\mathbb{R}^3)$, $m > 5/2$, generates a unique global classical solution, smooth for positive time, and every Leray–Hopf solution with that datum agrees with it.

The proof of [Theorem 1.1](#) is given in [Section 13](#). Its logical content is the conversion of qualitative material rigidity into a uniform price for target-retaining windows.

Remark 1.2 (Status of the phase-I manuscript). The material pullback identities, moving-cutoff monotonicity theorem, zero-production rigidity, and compactness argument are proved in this paper. The companion trilogy supplies the coherent critical orbit and discharges the covariant no-loss bridge through its exhaustive branch closures and canonical moving carrier. Pressure and chart motion are thus flux components of the same expenditure ledger rather than separate entropy corrections. The covariant heat-scale identity and strain–Gaussian completion are proved independently and describe the equality geometry. Appendices [C–G](#) state the replacements needed for a self-contained proof. Until those contracts are completed, [Theorem 1.1](#) is a short covariant reformulation of the companion trilogy, conditional only on the explicitly marked bridges.

Dependency map.

Input	Phase-I status	Independent destination
Weak covariant calculus	proved here	Appendix A
Material stretching–diffusion identities	proved for smooth flow	Appendix B
Canonical critical orbit	bridge assembled from trilogy precursors	Appendix C
Gauge, pressure, and adjoint control	partly proved; temporary local inputs	Appendix D
Covariant no-loss cascade	discharged by the Phase-I carrier bridge and closures	Appendices F–G
Phase-I moving-carrier realization	proved modulo trilogy routing	Appendix E
Equality-state exclusion	constant case proved here	Appendix G

Trilogy source ledger. The companion manuscripts are used only through the following precursor results. The rightmost column records what is still new in this paper.

Input	Precise source	New obligation here
Velocity concentration	[3, Corollary 6.8]	Galilean-invariant oscillation retention
Local Type I extraction	[3, Theorem 9.4 and Lemma 9.7]	Placement in one coherent orbit
Retained recurrent core	[2, Lemmas 10.18–10.20]	Infinite ordered windows without importing chain exclusion
Repaired chart and pressure	[4, Theorems 4.25, 4.30, and 4.35–4.37]	Galilean-invariant slice, global cocycle, and nondegenerate continuation
Compact retained windows	[4, Proposition 7.14 and Theorems 7.22, 9.16]	Compactness for the fixed oscillation observable and adjoint probes
Moving carrier and pressure tails	[4, Lemmas 8.114–8.116, Corollary 8.117, and Theorems 10.11 and 10.19]	One coherent schedule on the selected orbit
Scale and cost routing	[4, Corollaries 8.126 and 8.131 and Theorems 10.42 and 10.118]	Verification that every surviving branch has summable carrier error
Compact-cylinder Caccioppoli control	[4, Theorem 10.61]	Weak compactness on the retained carrier core

Neither the coherent-orbit bridge [Bridge hypothesis 9.2](#) nor the moving-carrier alternative [Bridge hypothesis 11.5](#) appears verbatim as a single theorem of the trilogy. The phase-I proof assembles the latter from the cited carrier estimates and named local closure theorems; the appendix records the exact synthesis to be proved independently.

2. SUITABLE SOLUTIONS AND THE SINGULAR TARGET

2.1. Suitable weak solutions. We use the standard finite-energy suitable class [7, 5, 8]. Thus

$$(2.1) \quad u \in L^\infty(0, T; L_\sigma^2(\mathbb{R}^3)) \cap L^2(0, T; \dot{H}_\sigma^1(\mathbb{R}^3)), \quad p \in L_{\text{loc}}^{3/2}(\mathbb{R}^3 \times (0, T)),$$

the equation holds in distributions, the global Leray–Hopf energy inequality holds, and the local energy inequality holds for every nonnegative compactly supported smooth test function. Here the subscript σ denotes divergence-free fields. A terminal point (x_0, T) is regular when u is locally bounded in a backward parabolic neighborhood of that point. We write

$$(2.2) \quad \Sigma_u(T) := \{x_0 \in \mathbb{R}^3 : (x_0, T) \text{ is not regular}\}.$$

For $z_0 = (x_0, t_0)$ and $r > 0$, write $Q_r(z_0) = B_r(x_0) \times (t_0 - r^2, t_0)$ and define

$$(2.3) \quad A(z_0, r) := r^{-1} \operatorname{ess\,sup}_{t_0 - r^2 < t < t_0} \int_{B_r(x_0)} |u(x, t)|^2 \, dx,$$

$$(2.4) \quad E(z_0, r) := r^{-1} \int_{Q_r(z_0)} |\nabla u|^2 \, dx \, dt,$$

$$(2.5) \quad C(z_0, r) := r^{-2} \int_{Q_r(z_0)} |u|^3 \, dx \, dt,$$

$$(2.6) \quad D(z_0, r) := r^{-2} \int_{t_0 - r^2}^{t_0} \int_{B_r(x_0)} |p - (p)_{B_r(x_0)}(t)|^{3/2} \, dx \, dt.$$

These quantities are invariant under the Navier–Stokes scaling. The pressure mean in D fixes the local pressure gauge.

Theorem 2.1 (Velocity concentration at a singular point). *There is a threshold $\varepsilon_v = \varepsilon_v(\nu) > 0$ such that every $x_0 \in \Sigma_u(T)$ admits radii $r_j \downarrow 0$ for which*

$$(2.7) \quad C((x_0, T), r_j) \geq \varepsilon_v \quad \text{for every } j.$$

PHASE-I SOURCE. For this draft, the velocity-only concentration formulation is imported from [3, Corollary 6.8]; its proof uses the standard suitable-solution compactness and velocity–pressure ε -regularity theory.

INDEPENDENT PROOF CONTRACT. Appendix C, Contract B.1, requires a direct derivation of (2.7) and the pressure normalization needed in the limiting argument.

2.2. Scale-critical concentration and local kinetic energy. The target of the variational interpretation is the set of germs retaining the velocity concentration forced by singularity.

Definition 2.2 (Solution-specific singular concentration target). Let $\varepsilon_v(\nu) > 0$ be the threshold in the velocity ε -regularity theorem and put $\eta_* := \varepsilon_v/2$. For a fixed solution u and terminal time T , let $\mathfrak{S}_{\text{sing},T}(u)$ be the set of pairs $(x_0, \{r_j\}_{j \geq 1})$, with $r_j \downarrow 0$, such that

$$(2.8) \quad C((x_0, T), r_j) \geq \eta_* \quad \text{for every } j.$$

Every $x_0 \in \Sigma_u(T)$ occurs in at least one element of $\mathfrak{S}_{\text{sing},T}(u)$ by Theorem 2.1. Conversely, if u is bounded in a backward neighborhood of (x_0, T) , then $C((x_0, T), r) = O(r^3)$, so no sequence based at x_0 belongs to $\mathfrak{S}_{\text{sing},T}(u)$.

The following estimate explains why localized kinetic probes are sufficient to measure the energy side of this target.

Lemma 2.3 (Local interpolation). *For every suitable weak solution and every interior cylinder $Q_r(z_0)$,*

$$(2.9) \quad C(z_0, r) \leq c(A(z_0, r)^{3/4} E(z_0, r)^{3/4} + A(z_0, r)^{3/2}),$$

where c is universal.

Proof. After translating the cylinder, the Sobolev and interpolation inequalities on B_r give, for almost every time,

$$\int_{B_r} |u|^3 \leq c \left(\int_{B_r} |u|^2 \right)^{3/4} \left(\int_{B_r} |\nabla u|^2 \right)^{3/4} + cr^{-3/2} \left(\int_{B_r} |u|^2 \right)^{3/2}.$$

Integrate over a time interval of length r^2 , apply Hölder's inequality to the first term, and divide by r^2 . Substitution of (2.3) and (2.4) yields (2.9). $\square$

Corollary 2.4 (Concentration requires store or expenditure). *Suppose $C(z_j, r_j) \geq \eta > 0$. Then*

$$\max\{A(z_j, r_j), E(z_j, r_j)\} \geq c\eta^{2/3} > 0.$$

If $E(z_j, r_j) \rightarrow 0$, then $\liminf_j A(z_j, r_j) \geq c\eta^{2/3}$.

Proof. Both assertions follow directly from (2.9). $\square$

Thus a singular target retains local kinetic energy, viscous expenditure, or both. The canonical orbit constructed in Section 9 sharpens this alternative by subtracting the weighted velocity mean. Pressure remains part of the local regularity criterion and of the conservative flux; viscosity supplies the dissipative current.

3. EULER TRANSPORT AND THE STOKES GRADIENT

3.1. The solenoidal Gelfand triple. Let

$$\mathcal{V}_\sigma := \dot{H}_\sigma^1(\mathbb{R}^3), \quad \mathcal{H}_\sigma := L_\sigma^2(\mathbb{R}^3), \quad \mathcal{V}_\sigma^* := \dot{H}_\sigma^{-1}(\mathbb{R}^3).$$

The Stokes operator $A_S : \mathcal{V}_\sigma \rightarrow \mathcal{V}_\sigma^*$ is defined weakly by

$$(3.1) \quad \langle A_S v, \varphi \rangle = \int_{\mathbb{R}^3} \nabla v : \nabla \varphi \, dx.$$

On smooth fields, $A_S = -\mathbb{P}\Delta$. Put

$$(3.2) \quad K := \nu A_S.$$

This is the Riesz isomorphism associated with the Dirichlet form multiplied by ν . For $g, h \in \mathcal{V}_\sigma^*$, define

$$(3.3) \quad (g, h)_{K^{-1}} := \langle g, K^{-1}h \rangle, \quad \|g\|_{K^{-1}}^2 = (g, g)_{K^{-1}}.$$

The projected convection operator is

$$(3.4) \quad B(v, w) := \mathbb{P}((v \cdot \nabla)w), \quad \langle B(v, w), \varphi \rangle = - \int_{\mathbb{R}^3} v \otimes w : \nabla \varphi \, dx.$$

For sufficiently regular divergence-free v ,

$$(3.5) \quad \langle B(v, v), v \rangle = 0.$$

Definition 3.1 (Euler-covariant derivative). Along a divergence-free curve $u(t)$, define

$$(3.6) \quad \mathfrak{D}_t^E u := \partial_t u + B(u, u).$$

This is the material derivative after projecting away the pressure constraint.

Proposition 3.2 (Exact covariant gradient identity). *Let u be a smooth solution of (1.1). For $\mathcal{E}(u) = \frac{1}{2}\|u\|_2^2$,*

$$(3.7) \quad \mathfrak{D}_t^E u = -K D\mathcal{E}(u) = -Ku.$$

If $J_\nu := -Ku$, then

$$(3.8) \quad \|J_\nu\|_{K^{-1}}^2 = \langle Ku, u \rangle = \nu \|\nabla u\|_2^2,$$

and

$$(3.9) \quad \frac{d}{dt} \mathcal{E}(u(t)) = -\|J_\nu(t)\|_{K^{-1}}^2.$$

Proof. Apply $\mathbb{P}$ to (1.1). Since $\mathbb{P}\nabla p = 0$, this gives $\partial_t u + B(u, u) + Ku = 0$, which is (3.7). The definition of the inverse metric gives

$$\|Ku\|_{K^{-1}}^2 = \langle Ku, u \rangle = \nu \int_{\mathbb{R}^3} |\nabla u|^2 \, dx.$$

Pairing (3.7) with u and using (3.5) proves (3.9). $\square$

3.2. The finite energy budget. For a Leray–Hopf solution, choose its standard weakly continuous representative and set

$$(3.10) \quad M(t) := \mathcal{E}(u(t)) + \nu \int_0^t \|\nabla u(s)\|_2^2 ds.$$

The global energy inequality says that M is nonincreasing after changing it on a null set. Its distributional derivative therefore defines a nonnegative Radon measure.

Theorem 3.3 (Energy expenditure for suitable weak solutions). *There is a nonnegative measure μ_{en} on $(0, T)$ such that, at every pair of times $0 \leq t_1 < t_2 < T$ for which the strong energy inequality is represented,*

$$(3.11) \quad \mathcal{E}(u(t_2)) + \int_{t_1}^{t_2} \|J_\nu(t)\|_{K^{-1}}^2 dt + \mu_{\text{en}}((t_1, t_2]) = \mathcal{E}(u(t_1)).$$

In particular,

$$(3.12) \quad \int_0^T \|J_\nu(t)\|_{K^{-1}}^2 dt \leq \mathcal{E}(u(0)).$$

For every classical solution, $\mu_{\text{en}} = 0$ on each compact interval of classical existence.

Proof. Define $\mu_{\text{en}} := -D_t M$ in distributions. Monotonicity of M gives $\mu_{\text{en}} \geq 0$, and the increment formula for M is exactly (3.11). The bound (3.12) follows by discarding the two nonnegative terminal terms. Smooth solutions satisfy the energy equality (3.9), so their defect measure vanishes. $\square$

Equation (3.11) is the finite-budget principle used in the rest of the paper. The available quantity is the initial kinetic energy, and the price of irreversible motion is the Stokes action $\|J_\nu\|_{K^{-1}}^2$.

4. MATERIAL DEFORMATION AND STRETCHING–DIFFUSION DUALITY

Let u be smooth on a time interval and let $X(a, t)$ be its material flow,

$$(4.1) \quad \partial_t X(a, t) = u(X(a, t), t), \quad X(a, t_0) = a.$$

Write

$$(4.2) \quad \mathbf{F} := \nabla_a X, \quad \mathbf{g} := \mathbf{F}^T \mathbf{F}, \quad \mathbf{\Omega}(a, t) := \omega(X(a, t), t), \quad \omega := \nabla \times u.$$

The tensor $\mathbf{g}$ is the pullback of the Euclidean metric by the volume-preserving map X . Denote the Eulerian strain by $S = \frac{1}{2}(\nabla u + \nabla u^T)$ and put $S_L = S \circ X$.

Proposition 4.1 (Exact material pullback). *For every smooth solution and every time on which (4.1) is defined,*

$$(4.3) \quad \det \mathbf{F} = 1, \quad \partial_t \mathbf{F} = (\nabla u \circ X) \mathbf{F},$$

$$(4.4) \quad \partial_t \mathbf{g} = 2\mathbf{F}^T S_L \mathbf{F}, \quad \det \mathbf{g} = 1.$$

For every smooth scalar field q ,

$$(4.5) \quad (\Delta_x q) \circ X = \partial_{a_i} (\mathbf{g}^{ij} \partial_{a_j} (q \circ X)),$$

where $(\mathbf{g}^{ij}) = \mathbf{g}^{-1}$. The pulled-back vorticity satisfies

$$(4.6) \quad \partial_t \mathbf{\Omega} = S_L \mathbf{\Omega} + \nu \partial_{a_i} (\mathbf{g}^{ij} \partial_{a_j} \mathbf{\Omega}).$$

Equivalently, the viscous Cauchy invariant $\xi := \mathbf{F}^{-1} \mathbf{\Omega}$ satisfies

$$(4.7) \quad \partial_t \xi = \nu \mathbf{F}^{-1} \partial_{a_i} (\mathbf{g}^{ij} \partial_{a_j} (\mathbf{F} \xi)).$$

Proof. Differentiating (4.1) in a gives the evolution of F . Jacobi's formula and incompressibility give

$$\partial_t \det F = (\operatorname{div} u) \circ X \det F = 0,$$

and the initial determinant is one. Differentiating $\mathbf{g} = F^T F$ proves (4.4). Formula (4.5) is the Laplace–Beltrami formula with volume density $\sqrt{\det \mathbf{g}} = 1$.

The vorticity equation is

$$(\partial_t + u \cdot \nabla) \omega = (\omega \cdot \nabla) u + \nu \Delta \omega.$$

The antisymmetric part of ∇u annihilates ω , so $(\omega \cdot \nabla) u = S\omega$. Composition with X and (4.5) give (4.6). Finally, $\partial_t F^{-1} = -F^{-1}(\nabla u \circ X)$; the inviscid terms cancel after substituting $\Omega = F\xi$, which proves (4.7). $\square$

Corollary 4.2 (Material enstrophy ledger). *If the fields decay sufficiently at infinity, then*

$$(4.8) \quad \frac{1}{2} \frac{d}{dt} \int_{\mathbb{R}^3} |\Omega|^2 da + \nu \int_{\mathbb{R}^3} g^{ij} \partial_{a_i} \Omega \cdot \partial_{a_j} \Omega da = \int_{\mathbb{R}^3} \Omega \cdot S_L \Omega da.$$

Proof. Pair (4.6) with Ω , integrate in a , and integrate the divergence term by parts. The material volume element is $da = dx$ because $\det F = 1$. $\square$

The duality in (4.6) is algebraic. If the singular values of F are $\sigma_1 \geq \sigma_2 \geq \sigma_3 > 0$, then

$$(4.9) \quad \sigma_1 \sigma_2 \sigma_3 = 1, \quad \lambda(\mathbf{g}^{-1}) = \{\sigma_1^{-2}, \sigma_2^{-2}, \sigma_3^{-2}\}, \quad \sigma_3^{-2} \geq \sigma_1.$$

Thus strong material stretching produces a strongly diffusive compressed direction. This observation alone is not a coercive estimate: the pulled-back vorticity could avoid variation in that direction. The equality mechanism is therefore geometric. Avoiding the enhanced direction requires simultaneous alignment of vorticity, strain, and the material eigenframe. The central rigidity problem is to show that a target-retaining orbit cannot sustain this alignment indefinitely unless it approaches a stretching-depleted, lower-dimensional state.

Definition 4.3 (Materially reduced state). A compact material limit is *materially reduced* on the retained core when its zero-expenditure equations determine a volume-preserving invariant distribution on which the pulled-back vorticity evolves without transverse variation, and the induced equation belongs to an explicitly classified regular lower-dimensional or shear class. Material reduction is a closure alternative for a terminal singular branch; it does not assert that a regular reduced velocity is spatially constant on a unit normalized window. In the phase-I carrier realization below, the stronger zero-expenditure equation $\nabla V = 0$ holds on the retained core and gives the oscillation-removable equality state of Definition 11.8.

The first sentence of Definition 4.3 is a proof obligation, not an automatic consequence of (4.9). The independent classification required to justify it is stated in Appendix G. This separation prevents the geometric interpretation from being used as an unproved regularity criterion.

5. COVARIANCE UNDER SCALE–TRANSLATION CHARTS

5.1. The repaired equation. Let $X(\tau) \in \mathbb{R}^3$ and $\Lambda(\tau) > 0$ be absolutely continuous, and define

$$(5.1) \quad x = X(\tau) + \Lambda(\tau)y, \quad \frac{dt}{d\tau} = \Lambda(\tau)^2,$$

$$(5.2) \quad u(x, t) = \Lambda(\tau)^{-1} V(y, \tau), \quad p(x, t) = \Lambda(\tau)^{-2} P(y, \tau).$$

Put

$$(5.3) \quad a(\tau) = \frac{\Lambda'(\tau)}{\Lambda(\tau)}, \quad b(\tau) = \frac{X'(\tau)}{\Lambda(\tau)}.$$

Proposition 5.1 (Covariant equation in a repaired chart). *The physical equation (1.1) is equivalent, in distributions on the image of the chart, to*

$$(5.4) \quad \partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a(V + y \cdot \nabla V) + b \cdot \nabla V, \quad \operatorname{div} V = 0.$$

Consequently the chart-covariant derivative

$$(5.5) \quad \mathfrak{D}_\tau^{X,\Lambda} V := \partial_\tau V + B(V, V) - a(V + y \cdot \nabla V) - b \cdot \nabla V$$

satisfies

$$(5.6) \quad \mathfrak{D}_\tau^{X,\Lambda} V = -\nu A_S V.$$

Proof. The change of variables in (5.1)–(5.2) gives the same factor Λ^{-3} in every term of the physical equation. After removing that factor, the time derivative contributes $a(V + y \cdot \nabla V) + b \cdot \nabla V$, giving (5.4). Applying the Helmholtz projection gives (5.6). $\square$

For the centered Type I variables

$$t = -e^{-\tau}, \quad \Lambda(\tau) = e^{-\tau/2}, \quad X = 0,$$

one has

$$(5.7) \quad a = -\frac{1}{2}, \quad b = 0,$$

and hence

$$(5.8) \quad \partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V - \frac{1}{2}(V + y \cdot \nabla V).$$

This is the convention of the Type I setup paper. In the repaired Type II charts, the companion representation yields

$$(5.9) \quad a, b \in L_{\text{loc}}^1(d\tau).$$

Bounded modulation is a separate hypothesis of the compact-window subclasses where the companion estimates require it.

5.2. Covariant energy and pressure. On a whole-space chart, physical kinetic energy and action take the form

$$(5.10) \quad \mathcal{E}(u(t(\tau))) = \frac{\Lambda(\tau)}{2} \int_{\mathbb{R}^3} |V(y, \tau)|^2 dy,$$

$$(5.11) \quad \|J_\nu(t)\|_{K^{-1}}^2 dt = \Lambda(\tau) \nu \int_{\mathbb{R}^3} |\nabla V(y, \tau)|^2 dy d\tau.$$

Proposition 5.2 (Energy identity in moving coordinates). *For a smooth whole-space represented solution,*

$$(5.12) \quad \frac{d}{d\tau} \left(\frac{\Lambda}{2} \|V\|_2^2 \right) = -\Lambda \nu \|\nabla V\|_2^2.$$

Thus (3.9) and (5.12) are the same scalar identity in different coordinates.

Proof. Pair (5.4) with V . Convection, translation, and pressure give zero. Moreover,

$$\int_{\mathbb{R}^3} V \cdot (V + y \cdot \nabla V) dy = -\frac{1}{2} \|V\|_2^2.$$

Since $\Lambda' = a\Lambda$, the derivative of the prefactor in (5.10) cancels this dilation term. The remaining term is the right-hand side of (5.12). $\square$

On compact cylinders, derivatives of the cutoff produce convective, pressure, scale, and translation fluxes in the local energy inequality. Their exact form is recorded in Appendix D; these are the local terms placed explicitly in the moving error density (11.22).

PHASE-I SOURCE. The fixed- and moving-cutoff estimates currently used for compact repaired windows are [4, Theorems 10.11, 10.19, and 10.61].

INDEPENDENT PROOF CONTRACT. Appendix D, Contracts C.2 and C.4, require derivation of the same estimates from the suitable local energy inequality and the canonical tail schedule.

Pressure remains the multiplier for incompressibility. On an inner ball use the decomposition

$$(5.13) \quad P = P_{\text{loc}} + H, \quad -\Delta P_{\text{loc}} = \partial_i \partial_j (\zeta V_i V_j), \quad \Delta H = 0,$$

with a spatial mean fixing H . The harmonic term can depend on space and time.

PHASE-I SOURCE. The local Calderón–Zygmund and harmonic bounds are temporarily imported from [4, Theorem 4.25].

INDEPENDENT PROOF CONTRACT. They are the required output of Appendix D, Contract C.2, with constants uniform on the normalized compact orbit.

Proposition 5.3 (Same-current covariance). *Let two admissible scale–translation charts represent the same physical solution on a common cylinder. The fields, test weights, and Stokes pairings transform by the corresponding pullback, while the physical current*

$$(5.14) \quad J_\nu = -\nu A_S u$$

is unchanged. In particular, (5.11) is invariant after the time change. Optimizing a concentration probe over admissible charts changes the probe but does not change J_ν .

Proof. Both charts are obtained from (5.1) by composing translations and positive dilations. The tensorial change of variables gives (5.10)–(5.11) in each chart. Since both right-hand sides equal the same physical energy and physical dissipation, their values agree. The current (5.14) is determined by u before either chart is selected. $\square$

6. SCALE-CRITICAL QUOTIENT GEOMETRY

The physical energy identity and the scale-critical concentration problem live at different levels. This section separates them and fixes the quotient structure used by the critical orbit.

6.1. Physical action and critical action. The Navier–Stokes scaling based at (x_0, t_0) is

$$(6.1) \quad u^{(\lambda)}(y, s) = \lambda u(x_0 + \lambda y, t_0 + \lambda^2 s), \quad p^{(\lambda)}(y, s) = \lambda^2 p(x_0 + \lambda y, t_0 + \lambda^2 s).$$

The quantities C, D are invariant under (6.1). The localized Dirichlet quantity is invariant after the parabolic factor is included:

$$(6.2) \quad r^{-1} \int_{Q_r(z_0)} |\nabla u|^2 dx dt = \int_{Q_1} |\nabla u^{(r)}|^2 dy ds.$$

This is the dissipation seen by a normalized unit window.

By contrast, the physical Stokes action transforms according to (5.11) and carries the factor Λ :

$$(6.3) \quad \int_{t(\tau_1)}^{t(\tau_2)} \nu \|\nabla u(t)\|_2^2 dt = \int_{\tau_1}^{\tau_2} \Lambda(\tau) \nu \|\nabla V(\tau)\|_2^2 d\tau.$$

Consequently, finite physical action does not control

$$(6.4) \quad \int_{\tau_1}^{\tau_2} \nu \int_{B_R} |\nabla V|^2 dy d\tau$$

uniformly along a collapsing orbit. For example, in centered Type I coordinates $\Lambda = e^{-\tau/2}$. A represented orbit with order-one Dirichlet expenditure on every unit interval has finite contribution to (6.3) but infinite contribution to (6.4). The global energy inequality alone therefore cannot supply the required contradiction.

The corrected moving-carrier entropy is designed precisely at the level of (6.4). Its lower bound replaces the missing unweighted global budget in represented time. The production gap then says that a retained core must spend a fixed amount of this critical action, or of the other covariant production terms, on every bad window.

6.2. Galilean and pressure quotients. For a constant vector $c \in \mathbb{R}^3$, the Galilean transform (6.5)

$$u^c(x, t) := u(x - ct, t) + c, \quad p^c(x, t) := p(x - ct, t)$$

preserves (1.1). Addition of a function $\pi(t)$ to the pressure also preserves the equation. These modes have no bearing on regularity but can change an uncentered local kinetic integral. The variance (8.6) is invariant under the induced constant-velocity action:

$$(6.6) \quad \frac{1}{2} \int m |(V + c) - \overline{(V + c)}_m|^2 = \frac{1}{2} \int m |V - \bar{V}_m|^2.$$

For $c \neq 0$, (6.5) does not preserve the global finite-energy class on $\mathbb{R}^3$. It is used only as a symmetry of local profiles and observables; the physical finite-energy solution and its Stokes current are never replaced by u^c . Likewise, D and the pressure atlas depend only on the pressure class modulo functions of time.

Fix nonzero nonnegative radial cutoffs $\zeta_0, \zeta_1 \in C_c^\infty(B_R)$, a level $\Theta_0 > 0$, and write $V_{\lambda, q}(y) = \lambda V(q + \lambda y)$. For a nonzero cutoff ξ , set

$$(6.7) \quad \langle W \rangle_\xi := \frac{\int \xi W \, dy}{\int \xi \, dy}, \quad W_\xi^\circ := W - \langle W \rangle_\xi.$$

Translation and dilation are fixed by the four-component, Galilean-invariant gauge map

$$(6.8) \quad \begin{aligned} \mathfrak{F}_0(\lambda, q; V) &:= \int \zeta_0 |(V_{\lambda, q})_{\zeta_0}^\circ|^3 \, dy - \Theta_0, \\ \mathfrak{F}_j(\lambda, q; V) &:= \int y_j \zeta_1 |(V_{\lambda, q})_{\zeta_1}^\circ|^2 \, dy, \quad 1 \leq j \leq 3. \end{aligned}$$

Indeed, replacing V by $V + c$ adds the constant λc to $V_{\lambda, q}$, which is removed by both means in (6.7). Thus the gauge selects only scale and spatial center; it does not select a velocity representative. At a zero for which

$$(6.9) \quad D_{(\lambda, q)} \mathfrak{F}(1, 0; V) \in \mathbb{R}^{4 \times 4} \quad \text{is invertible,}$$

the implicit-function theorem selects the scale and center locally. Differentiating the four constraints produces (a, b) . Degeneration of (6.9) is an explicit alternative in the critical-orbit bridge; no smooth continuation through a degenerate point is assumed.

6.3. The quotient topology. On a fixed represented cylinder $B_R \times I$, put

$$(6.10) \quad \mathcal{X}_R(I) := (L_I^\infty L^2(B_R) \cap L_I^2 H^1(B_R) \cap L^3(B_R \times I)) \times L^{3/2}(B_R \times I) \times L^1(I; \mathbb{R}^4),$$

where the factors represent V , P , and (a, b) . The compactness topology consists of strong local L^3 convergence for velocity, weak local $L^2 H^1$ convergence for its gradient, strong local $L^{3/2}$ convergence for the normalized pressure, and L^1 convergence of modulation coefficients on the selected windows. Adjoint densities and cutoffs are added with the topology in (A.7) and (A.6).

Let $\mathcal{G}_{\text{ch}}$ denote the local chart action generated by translations, positive dilations, and pressure gauge. Near a zero satisfying (6.9), the normalized slice (9.3), together with (9.4), intersects the

corresponding $\mathcal{G}_{\text{ch}}$ -orbit locally and transversely. Global continuation and consistency of these local slices are part of [Bridge hypothesis 9.2](#). Constant-velocity Galilean symmetry is not fixed by a time-dependent mean condition; instead, every scalar used below is invariant under that symmetry by construction. A compact family of normalized windows may therefore be studied on the chart slice and then quotiented by the residual Galilean action.

Proposition 6.1 (Covariance of the reduced data). *On the nondegenerate normalized slice, the following objects depend only on the quotient state:*

- (i) *the oscillatory kinetic energy $\mathcal{E}_\Gamma^\circ$ and concentration $\mathcal{C}_\Gamma^\circ$;*
- (ii) *the physical Stokes current and its inverse-metric norm;*
- (iii) *the ray distance after the target probe is pulled back with the chart;*
- (iv) *the normalized pressure class and, for a fixed padded terminal problem, the localized adjoint probability measure $m \, dy$.*

Proof. The first item follows from change of variables and (6.6). The second and third follow from [Proposition 5.3](#) and covariance of orthogonal projection in the Stokes metric. Pressure means are removed by (9.4). For a fixed padded terminal problem, the normalized adjoint equation pulls back covariantly when its domain, cutoff, and terminal datum are pulled back with the chart, and normalization fixes the mass. No comparison between distinct terminal problems is asserted or needed for the moving-carrier entropy. $\square$

The proposition does not supply compactness or a summable carrier schedule. Those are the contents of [Bridge hypothesis 9.2](#) and [Bridge hypothesis 11.5](#), respectively.

7. BACKWARD ADJOINT LOCALIZATIONS AND ENTROPY

The adjoint density supplies scale-relative localization. Its entropy is kept separate from the kinetic-energy gradient identity; this separation prevents pressure and critical norms from acquiring artificial variational equations.

7.1. The normalized finite-horizon adjoint problem. Fix a repaired cylinder $B_R \times (\tau_0, \sigma)$, let

$$(7.1) \quad c(y, \tau) := V(y, \tau) - a(\tau)y - b(\tau), \quad \operatorname{div} c = -3a,$$

and choose $\chi \in C_c^\infty(B_R)$, $\chi \geq 0$, together with a nonnegative terminal datum $h \in C_c^\infty(B_R)$, $h \not\equiv 0$. First solve the linear terminal-boundary problem

$$(7.2) \quad -\partial_\tau \tilde{\rho} - \nu \Delta \tilde{\rho} - \operatorname{div}(c \tilde{\rho}) = 0, \quad \tilde{\rho}|_{\partial B_R} = 0, \quad \tilde{\rho}(\sigma) = h.$$

Define

$$(7.3) \quad Z(\tau) := \int_{B_R} \chi^2 \tilde{\rho} \, dy, \quad \rho := \frac{\tilde{\rho}}{Z}, \quad m := \chi^2 \rho.$$

We work on intervals on which $Z > 0$. Then $m \geq 0$ and

$$(7.4) \quad \int_{B_R} m(y, \tau) \, dy = 1.$$

For a retained unit window $I = [s, s+1]$, the adjoint problem is always solved on a padded interval ending at

$$(7.5) \quad \sigma_I = s + 1 + \kappa_*, \quad \kappa_* > 0.$$

Consequently

$$(7.6) \quad \kappa_* e^{-2M_*} \leq \theta_I(\tau) \leq (1 + \kappa_*) e^{2M_*} \quad (\tau \in I).$$

Here M_* is a uniform bound for $\int_s^{\sigma_I} |a(r)| dr$ on the normalized padded windows, and θ_I is the unique solution of the covariant terminal-value problem

$$(7.7) \quad \theta'_I + 2a\theta_I = -1, \quad \theta_I(\sigma_I) = 0.$$

Indeed,

$$(7.8) \quad \theta_I(\tau) = \int_\tau^{\sigma_I} \exp\left(2 \int_\tau^r a(q) dq\right) dr,$$

which proves (7.6). The Gaussian entropy is evaluated only on the retained window, where $\theta_I > 0$, not at the terminal time. It is a covariant diagnostic on each padded window; the global monotonicity ledger constructed below does not compare or reset finite-horizon adjoint densities.

Lemma 7.1 (Normalized adjoint equation). *Set*

$$(7.9) \quad \kappa(\tau) := \frac{Z'(\tau)}{Z(\tau)} = \int_{B_R} (-\nu \Delta \chi^2 + c \cdot \nabla \chi^2) \rho dy.$$

Then

$$(7.10) \quad -\partial_\tau \rho - \nu \Delta \rho - \operatorname{div}(c\rho) = \kappa \rho$$

in distributions, and (7.4) holds at every time at which the weak solution is represented continuously in L^2 .

Proof. Differentiate Z using (7.2) and integrate by parts. The support of χ is contained in B_R , so

$$Z' = \int_{B_R} (-\nu \Delta \chi^2 + c \cdot \nabla \chi^2) \tilde{\rho} dy.$$

Division by Z gives (7.9). Substituting $\tilde{\rho} = Z\rho$ into (7.2) gives (7.10). The last statement follows from the definition of Z . $\square$

This construction solves the linear equation before normalization. The coefficient κ is therefore determined by the solution and does not enter the linear terminal problem.

Lemma 7.2 (Weak adjoint construction). *Assume*

$$(7.11) \quad V \in L_\tau^2 H_y^1(B_R), \quad a, b \in L_\tau^1, \quad \operatorname{div} c = -3a.$$

Then (7.2) has a nonnegative weak solution with

$$\tilde{\rho} \in L_\tau^\infty L_y^2(B_R) \cap L_\tau^2 H_{0,y}^1(B_R).$$

It is obtained as the weak limit of solutions with smooth drifts c_ε . On every compact subinterval where $Z \geq z_0 > 0$, the normalized densities satisfy the same bounds, and $\sqrt{\rho} \in L_\tau^2 H_{\operatorname{loc},y}^1$ whenever their Fisher information is uniformly bounded.

Proof. Reverse time in (7.2) and approximate V, a, b by smooth functions preserving $\operatorname{div} c_\varepsilon = -3a_\varepsilon$. Galerkin approximation gives smooth weak solutions. Testing by $\tilde{\rho}_\varepsilon$ makes the transport contribution $-\frac{1}{2} \int (\operatorname{div} c_\varepsilon) |\tilde{\rho}_\varepsilon|^2$, so Gronwall's inequality uses only $a_\varepsilon \in L^1$. This yields uniform $L^\infty L^2 \cap L^2 H_0^1$ bounds. Choose the approximating drifts to converge strongly in $L^2 L^6 + L^1 L^\infty$. The first summand follows from $V \in L^2 H^1$, and the affine part belongs to the second summand on B_R . The bounds on $\tilde{\rho}_\varepsilon$ then allow $c_\varepsilon \tilde{\rho}_\varepsilon$ to pass to the limit against every smooth test function. Weak compactness gives a solution of (7.2); testing the negative part proves nonnegativity. Division by $Z \geq z_0$ preserves the estimates. Finally,

$$\int \frac{|\nabla \rho|^2}{\rho + \varepsilon} dy = 4 \int |\nabla \sqrt{\rho + \varepsilon}|^2 dy,$$

and weak lower semicontinuity gives the last assertion as $\varepsilon \downarrow 0$. $\square$

7.2. Transport identity and the Gaussian equality model. For smooth positive ρ , write

$$(7.12) \quad \rho = (4\pi\nu\theta)^{-3/2}e^{-f}, \quad \theta > 0.$$

Equation (7.10) gives

$$(7.13) \quad \partial_\tau f + \nu\Delta f - \nu|\nabla f|^2 + c \cdot \nabla f = \kappa - \frac{3\theta'}{2\theta} + \operatorname{div} c.$$

Lemma 7.3 (Localized adjoint transport identity). *For every smooth scalar F ,*

$$(7.14) \quad \begin{aligned} \frac{d}{d\tau} \int_{B_R} \chi^2 F \rho \, dy &= \int_{B_R} \chi^2 (\partial_\tau F - \nu\Delta F + c \cdot \nabla F) \rho \, dy \\ &\quad + \int_{B_R} [F(-\nu\Delta\chi^2 + c \cdot \nabla\chi^2) - 2\nu\nabla\chi^2 \cdot \nabla F] \rho \, dy \\ &\quad - \kappa \int_{B_R} \chi^2 F \rho \, dy. \end{aligned}$$

Proof. Differentiate the integral, substitute (7.10), and integrate the Laplacian and divergence terms by parts. All spatial boundary terms vanish because $\chi \Subset B_R$. $\square$

The three lines of (7.14) separate interior evolution, cutoff transport, and normalization. Taking

$$(7.15) \quad W_f := \nu\theta|\nabla f|^2 + f - 3, \quad \mathcal{W}_{\text{ad},\chi}(\tau) := \int_{B_R} \chi^2 W_f \rho \, dy,$$

gives the exact localized derivative by setting $F = W_f$ in (7.14) and using (7.13). In particular, derivatives of χ and the scalar κ remain visible in the last two lines. The velocity drift contributes through ∇c , and its compression contributes through $\operatorname{div} c = -3a$. No sign is assigned to those local terms without an additional estimate.

The equality model is most transparent without localization.

Proposition 7.4 (Adjoint entropy production with a variable heat scale). *Let ρ be a smooth positive probability density on $\mathbb{R}^3$ satisfying*

$$-\partial_\tau \rho - \nu\Delta \rho - \operatorname{div}(c\rho) = 0$$

with sufficient decay. With f as in (7.12), for every positive absolutely continuous θ one has

$$(7.16) \quad \begin{aligned} \frac{d}{d\tau} \int_{\mathbb{R}^3} (\nu\theta|\nabla f|^2 + f - 3)\rho \, dy &= 2\nu^2\theta \int_{\mathbb{R}^3} \left| \nabla^2 f - \frac{1}{2\nu\theta} I \right|^2 \rho \, dy \\ &\quad + \int_{\mathbb{R}^3} (\operatorname{div} c)\rho \, dy \\ &\quad + 2\nu\theta \int_{\mathbb{R}^3} (\nabla f \cdot \nabla \operatorname{div} c - (\nabla f)^T (\nabla c) \nabla f) \rho \, dy \\ &\quad + (\theta' + 1) \left(\nu \int_{\mathbb{R}^3} |\nabla f|^2 \rho \, dy - \frac{3}{2\theta} \right). \end{aligned}$$

For $c = 0$ and $\theta' = -1$, the derivative is nonnegative and vanishes precisely when $\nabla^2 f = (2\nu\theta)^{-1}I$ on the support of ρ .

Proof. Let $H(\rho) = -\int \rho \log \rho$ and $I(\rho) = \int |\nabla \log \rho|^2 \rho$. Direct differentiation with the backward adjoint equation and integration by parts give

$$\frac{dH}{d\tau} = -\nu I + \int (\operatorname{div} c)\rho$$

and

$$\frac{dI}{d\tau} = 2\nu \int |\nabla^2 \log \rho|^2 \rho - 2 \int (\nabla \log \rho \cdot \nabla \operatorname{div} c + (\nabla \log \rho)^T (\nabla c) \nabla \log \rho) \rho.$$

Differentiate

$$\nu \theta I + H - \frac{3}{2} \log(4\pi\nu\theta) - 3$$

in τ , use $\nabla \log \rho = -\nabla f$, and first collect the terms corresponding to $\theta' = -1$. Completing their square gives the first three lines of (7.16). The difference between general θ' and -1 is exactly its last line. $\square$

For the repaired Navier–Stokes drift $c = V - ay - b$, one has $\nabla \operatorname{div} c = 0$ and $\nabla c = \nabla V - aI$. Hence the choice (7.7) cancels every scalar scale term in (7.16) and gives

$$(7.17) \quad \begin{aligned} \frac{d}{d\tau} \mathcal{W}_{\text{ad}} &= 2\nu^2 \theta \int_{\mathbb{R}^3} \left| \nabla^2 f - \frac{1}{2\nu\theta} I \right|^2 \rho \, dy \\ &\quad - 2\nu\theta \int_{\mathbb{R}^3} (\nabla f)^T (\nabla V)_{\text{sym}} \nabla f \rho \, dy. \end{aligned}$$

Here $\mathcal{W}_{\text{ad}}$ denotes the whole-space entropy in Proposition 7.4, and $(\nabla V)_{\text{sym}} := \frac{1}{2}(\nabla V + (\nabla V)^T)$. Formula (7.17) isolates the positive Gaussian square from velocity strain. The translation coefficient b drops out because it has no spatial derivative, while a drops out because the heat scale is transported by (7.7). The strain term has no sign by itself, but it completes a square with viscous production below.

In the unmodulated case $a = b = 0$, for which $\theta' = -1$, the Euclidean heat kernel is

$$(7.18) \quad \rho_*(y, \tau) = (4\pi\nu\theta)^{-3/2} \exp\left(-\frac{|y|^2}{4\nu\theta}\right), \quad f_* = \frac{|y|^2}{4\nu\theta},$$

and satisfies

$$\nabla^2 f_* = \frac{1}{2\nu\theta} I, \quad \int_{\mathbb{R}^3} (\nu\theta |\nabla f_*|^2 + f_* - 3) \rho_* \, dy = 0.$$

Hence both the entropy and its production vanish. This is the Euclidean normalization of Perelman’s $\mathcal{W}$ -entropy [10].

For weak densities, the Fisher term is defined by

$$(7.19) \quad \int \chi^2 \frac{|\nabla \rho|^2}{\rho} \, dy := 4 \int \chi^2 |\nabla \sqrt{\rho}|^2 \, dy,$$

with value $+\infty$ when $\sqrt{\rho} \notin H_{\text{loc}}^1$. The entropy uses $0 \log 0 = 0$. At weak regularity the Fisher information has its usual lower-semicontinuous extension. The Shannon term in $\mathcal{W}_{\text{ad}, \chi}$ has the opposite semicontinuity orientation, so its passage to a limit requires an explicit strong-convergence and uniform-integrability argument, as recorded in Contract C.3; no blanket lower-semicontinuity claim is made for the full entropy. No pointwise equation for $f = -\log \rho + \text{constant}$ is asserted on the zero set of ρ . Smooth positive adjoint approximations justify the localized transport formulae needed below.

8. VISCOUS AIMING AND THE RAY-ALIGNMENT DEFECT

8.1. Scale-critical kinetic probes. On a compact represented-time interval I_τ , an admissible probe consists of a repaired scale–translation chart (X, Λ) and a nonnegative function

$$(8.1) \quad \begin{aligned} m &\in W^{1,1}(I_\tau; W^{1,\infty}(B_R)) \cap L^\infty(I_\tau; W^{2,\infty}(B_R)), \\ \operatorname{supp} m(\cdot, \tau) &\subset B_{R_0} \quad (0 < R_0 < R), \quad \int_{B_R} m(y, \tau) \, dy = 1. \end{aligned}$$

The smooth adjoint weights $m = \chi^2 \rho$ from [Section 7](#), and regularized weak adjoint weights with the same normalization, are the principal examples. The physical multiplier is

$$(8.2) \quad w_\Gamma(x, t(\tau)) := \Lambda(\tau)^{-1} m \left(\frac{x - X(\tau)}{\Lambda(\tau)}, \tau \right).$$

Its physical mass and weighted velocity mean are

$$(8.3) \quad M_\Gamma(t) := \int_{\mathbb{R}^3} w_\Gamma dx = \Lambda(\tau)^2, \quad \bar{u}_\Gamma(t) := M_\Gamma(t)^{-1} \int_{\mathbb{R}^3} w_\Gamma u dx.$$

The Galilean-reduced localized kinetic probe is

$$(8.4) \quad F_\Gamma(u(t)) := \frac{1}{2} \int_{\mathbb{R}^3} w_\Gamma(x, t) |u(x, t) - \bar{u}_\Gamma(t)|^2 dx.$$

On the represented cylinder,

$$(8.5) \quad F_\Gamma(u(t(\tau))) = \frac{1}{2} \int_{B_R} m(y, \tau) |V(y, \tau) - \bar{V}_m(\tau)|^2 dy, \quad \bar{V}_m := \int_{B_R} m V dy.$$

Thus F_Γ is invariant under the Navier–Stokes scaling and is a Galilean-invariant version of the local quantity A . Denote its chart expression by

$$(8.6) \quad \mathcal{E}_\Gamma^\circ(V(\tau)) := \frac{1}{2} \int_{B_R} m |V - \bar{V}_m|^2 dy.$$

For variations of V with m fixed,

$$(8.7) \quad D\mathcal{E}_\Gamma^\circ(V)[\varphi] = \int_{B_R} m (V - \bar{V}_m) \cdot \varphi dy.$$

Indeed, the variation of $\bar{V}_m$ pairs with $\int m(V - \bar{V}_m) = 0$. Thus the target gradient below is the Stokes gradient of the Galilean-reduced variance in every representative. No time-dependent velocity-centering condition is imposed on the chart.

Let

$$(8.8) \quad q_\Gamma := \mathbb{P}(w_\Gamma(u - \bar{u}_\Gamma)).$$

For variations $\varphi \in \mathcal{H}_\sigma$,

$$DF_\Gamma(u)[\varphi] = (q_\Gamma, \varphi)_{L^2}.$$

Since $q_\Gamma \in \mathcal{V}_\sigma$ for almost every time, this derivative extends to currents $J \in \mathcal{V}_\sigma^*$ by $DF_\Gamma(u)[J] := \langle J, q_\Gamma \rangle$. The Stokes gradient current of the probe is

$$(8.9) \quad G_\Gamma := K q_\Gamma.$$

Define the viscous action, aiming energy, and signed target power by

$$(8.10) \quad \mathcal{A}_\nu(u) := \|J_\nu\|_{K^{-1}}^2 = \nu \|\nabla u\|_2^2,$$

$$(8.11) \quad \mathcal{G}_\Gamma(u) := \|G_\Gamma\|_{K^{-1}}^2 = \nu \|\nabla q_\Gamma\|_2^2,$$

$$(8.12) \quad \mathcal{P}_\Gamma(u) := (J_\nu, G_\Gamma)_{K^{-1}} = DF_\Gamma(u)[J_\nu].$$

Lemma 8.1 (Explicit viscous power). *For smooth u and w_Γ ,*

$$(8.13) \quad \mathcal{P}_\Gamma = -\nu \int_{\mathbb{R}^3} w_\Gamma |\nabla u|^2 dx + \frac{\nu}{2} \int_{\mathbb{R}^3} |u - \bar{u}_\Gamma|^2 \Delta w_\Gamma dx.$$

The first term is local viscous expenditure. The second is the diffusive flux across the spatial variation of the probe.

Proof. Since $A_S u$ is solenoidal and $\mathbb{P}$ is self-adjoint,

$$\mathcal{P}_\Gamma = -\nu \int \nabla u : \nabla q_\Gamma = -\nu \int \nabla u : \nabla (w_\Gamma(u - \bar{u}_\Gamma)).$$

Expand the last gradient and integrate $-\nu \int \partial_j u_i (u_i - \bar{u}_{\Gamma,i}) \partial_j w_\Gamma$ by parts. This gives (8.13). $\square$

In the represented variables, the power measure has the scale-free form

$$(8.14) \quad \mathcal{P}_\Gamma(u(t(\tau))) \, dt = \widehat{\mathcal{P}}_\Gamma(V(\tau)) \, d\tau, \quad \widehat{\mathcal{P}}_\Gamma(V) := -\nu \int_{B_R} m |\nabla V|^2 \, dy + \frac{\nu}{2} \int_{B_R} |V - \bar{V}_m|^2 \Delta m \, dy.$$

For smooth solutions the total derivative of the probe splits as

$$(8.15) \quad \frac{d}{dt} F_\Gamma(u(t)) = \mathcal{T}_\Gamma(u, p) + \mathcal{P}_\Gamma(u),$$

where

$$(8.16) \quad \mathcal{T}_\Gamma(u, p) := \frac{1}{2} \int |u - \bar{u}_\Gamma|^2 (\partial_t w_\Gamma + u \cdot \nabla w_\Gamma) \, dx + \int p (u - \bar{u}_\Gamma) \cdot \nabla w_\Gamma \, dx.$$

The term $\mathcal{T}_\Gamma$ contains conservative transport, pressure flux, and the motion of the scale-center coordinates. The term $\mathcal{P}_\Gamma$ contains viscosity alone. For suitable weak solutions, define

$$(8.17) \quad \mu_{\text{loc}} := - \left[\partial_t \frac{|u|^2}{2} + \operatorname{div} \left(\left(\frac{|u|^2}{2} + p \right) u \right) - \nu \Delta \frac{|u|^2}{2} + \nu |\nabla u|^2 \right].$$

The local energy inequality says precisely that this distribution is a nonnegative Radon measure. Combining it with the weak momentum identity for the weighted mean gives

$$(8.18) \quad F_\Gamma(t_2) - F_\Gamma(t_1) = \int_{t_1}^{t_2} (\mathcal{T}_\Gamma + \mathcal{P}_\Gamma) \, dt - \mu_{\text{loc}}(w_\Gamma; [t_1, t_2]),$$

where $\mu_{\text{loc}}(w_\Gamma; [t_1, t_2])$ denotes the measure in (8.17) paired with the nonnegative test weight on the indicated time interval. The identity holds at the corresponding local energy times; arbitrary compact subintervals follow by one-sided representatives.

8.2. Alignment and action-aiming.

Definition 8.2 (Viscous alignment). Set

$$(8.19) \quad \operatorname{Align}_\Gamma(u) := \begin{cases} \frac{(\mathcal{P}_\Gamma(u)_+)^2}{\mathcal{A}_\nu(u) \mathcal{G}_\Gamma(u)}, & \mathcal{A}_\nu(u) \mathcal{G}_\Gamma(u) > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Positive power means that the viscous current points toward increasing the localized kinetic probe. Negative power means that viscosity lowers that probe.

Theorem 8.3 (Action-aiming inequality). *For almost every time,*

$$(8.20) \quad (\mathcal{P}_\Gamma(u)_+)^2 \leq \mathcal{A}_\nu(u) \mathcal{G}_\Gamma(u), \quad 0 \leq \operatorname{Align}_\Gamma(u) \leq 1.$$

For every interval $I \subseteq (0, T)$,

$$(8.21) \quad \int_I \mathcal{P}_\Gamma(u(t))_+ \, dt \leq \left(\int_I \mathcal{A}_\nu(u(t)) \, dt \right)^{1/2} \left(\int_I \mathcal{G}_\Gamma(u(t)) \, dt \right)^{1/2}.$$

The first factor is at most $\mathcal{E}(u(0))^{1/2}$.

Proof. The definitions give

$$\mathcal{P}_\Gamma = (J_\nu, G_\Gamma)_{K^{-1}}.$$

Cauchy–Schwarz in the K^{-1} -metric proves (8.20). A second application of Cauchy–Schwarz in time proves (8.21). The energy bound is (3.12). $\square$

The inequality separates the amount of irreversible motion from its orientation. Equal action can produce zero or maximal positive power, depending on the angle with the target gradient.

Definition 8.4 (Covariant ray-alignment defect). For a sufficiently regular divergence-free comparison curve v , set

$$(8.22) \quad R(v) := \mathfrak{D}_t^E v + K v.$$

For an admissible probe Γ and interval I , define

$$(8.23) \quad \begin{aligned} \mathcal{D}_{\Gamma, I}^{\text{ray}}(v) &:= \frac{1}{2} \int_I \|R(v)\|_{K^{-1}}^2 dt \\ &\quad + \frac{1}{2} \int_I \left(\mathcal{A}_\nu(v) - \frac{(\mathcal{P}_\Gamma(v)_+)^2}{\mathcal{G}_\Gamma(v)} \right) dt, \end{aligned}$$

where the quotient is defined as zero when $\mathcal{G}_\Gamma = 0$.

Proposition 8.5 (Metric meaning of the ray defect). *The functional $\mathcal{D}_{\Gamma, I}^{\text{ray}}$ is nonnegative. Pointwise in time,*

$$(8.24) \quad \mathcal{A}_\nu - \frac{(\mathcal{P}_{\Gamma+})^2}{\mathcal{G}_\Gamma} = \text{dist}_{K^{-1}}^2(J_\nu, \{\alpha G_\Gamma : \alpha \geq 0\}).$$

For an exact Navier–Stokes solution, the equation-defect term in (8.23) vanishes. Equality $\mathcal{D}_{\Gamma, I}^{\text{ray}}(u) = 0$ then means that the actual viscous current lies on the nonnegative target-gradient ray almost everywhere on I . Moreover,

$$(8.25) \quad 2\mathcal{D}_{\Gamma, I}^{\text{ray}}(u) = \int_I \mathcal{A}_\nu(u(t))(1 - \text{Align}_\Gamma(u(t))) dt.$$

Proof. Orthogonal projection of a vector J onto the closed ray generated by a nonzero vector G has coefficient $(J, G)_+ / \|G\|^2$. Its squared residual norm is $\|J\|^2 - (J, G)_+^2 / \|G\|^2$. Apply this identity in the K^{-1} -metric. When $G = 0$, the distance is $\|J\|$, in agreement with the convention in (8.23). The equation-defect statement is (3.7). Substitution of (8.19) gives (8.25). $\square$

Let $\mathfrak{G}_{\text{adm}}(I)$ denote repaired probes with scale-relative cutoff bounds and normalized adjoint weights as in Section 7. Let $\mathfrak{G}_{\text{adm}}(t)$ be their time- t slices. Define

$$(8.26) \quad \text{Align}_*(t) := \sup_{\Gamma \in \mathfrak{G}_{\text{adm}}(t)} \text{Align}_\Gamma(t), \quad \mathcal{D}_*^{\text{ray}}(I) := \inf_{\Gamma \in \mathfrak{G}_{\text{adm}}(I)} \mathcal{D}_{\Gamma, I}^{\text{ray}}(u).$$

Suprema and infima are understood through approximating sequences; no attainment is required. The Stokes mobility K and current J_ν are fixed throughout these optimizations.

Theorem 8.6 (Local energy budget and alignment bound). *For every admissible probe and interval $[t_1, t_2] \subseteq (0, T)$, the following three statements hold simultaneously:*

- (i) *global energy decreases by viscous action and the nonnegative energy defect according to (3.11);*
- (ii) *the localized probe satisfies (8.18), separating coordinate/transport flux from viscous target power;*
- (iii) *positive viscous target power satisfies (8.21) and is therefore paid for by the finite global action.*

These conclusions remain valid along an optimizing sequence for (8.26).

Proof. Items (i)–(iii) are [Theorem 3.3](#), the suitable local energy inequality tested by w_Γ , and [Theorem 8.3](#), respectively. Each inequality holds for every admissible probe, so it holds termwise along any approximating sequence in (8.26). $\square$

8.3. Alignment, coercivity, and coherent probe selection. The alignment is an angle in the Stokes metric. It contains no information about the amount of action unless it is paired with $\mathcal{A}_\nu$, and it contains no compactness information unless the probe belongs to a normalized family. The following elementary observation fixes the equality orientation.

Lemma 8.7 (Positive-ray equality for a kinetic probe). *Let $w \geq 0$ have positive finite mass, put $\bar{u}_w = (\int w)^{-1} \int wu$, let $q = \mathbb{P}(w(u - \bar{u}_w))$, and suppose $u \in L_\sigma^2 \cap \dot{H}_\sigma^1$. If*

$$(8.27) \quad -Ku = \alpha Kq \quad \text{for some } \alpha \geq 0,$$

then $u = 0$.

Proof. Apply K^{-1} in the solenoidal Gelfand triple and pair the resulting identity $-u = \alpha q$ with u . Since u is solenoidal,

$$-\|u\|_2^2 = \alpha(q, u)_{L^2} = \alpha \int w|u - \bar{u}_w|^2 dx \geq 0.$$

Both sides vanish, so $u = 0$. $\square$

For a finite-energy whole-space state, vanishing ray defect is therefore a rigid equality. Along a singular rescaling, however, the physical ray defect is weighted by Λ exactly as in (6.3). It may have finite physical integral even when its scale-critical chart density remains positive on every unit window. This explains why the ray defect is only a covariant diagnostic; the exclusion proof instead controls the local Dirichlet chart density through the moving carrier.

Probe optimization also has a precise order. The physical solution first determines J_ν and the canonical chart. A fixed terminal-data rule then selects the finite-horizon adjoint problem on the padded window (7.5). Only after these choices are fixed may one take a supremum in Align or an infimum in $\mathcal{D}^{\text{ray}}$. Replacing the current by the gradient of an optimizing probe would change the equation and is not an admissible variation.

Definition 8.8 (Coherent probe cocycle). A coherent probe cocycle on a represented orbit is a family $\{\Gamma_j\}_{j \geq 1}$ on padded retained windows such that:

- (i) every Γ_j is constructed from the same physical solution by a fixed terminal-data rule and has the buffer (7.6);
- (ii) chart transitions satisfy the cocycle law in [Bridge hypothesis 9.2](#);
- (iii) the current paired with every target is the pullback of $J_\nu = -\nu A_S u$;
- (iv) approximating probes converge in the compact topology required by the local adjoint construction.

The terminal adjoint equation is a testing device. Coherence here means that all probes test the same physical current. It does not identify distinct finite-horizon adjoint densities with one global density, and no such identification is used by the carrier entropy below.

Proposition 8.9 (Scale-critical window form of the ray defect). *Let I_τ be a represented interval and I_t its physical image. For an exact Navier–Stokes solution, the pullback of the physical ray defect has the form*

$$(8.28) \quad 2\mathcal{D}_{\Gamma, I_t}^{\text{ray}}(u) = \int_{I_\tau} \Lambda(\tau) \widehat{\mathfrak{D}}_{\text{ray}, \Gamma}(V(\tau)) d\tau,$$

where $\widehat{\mathfrak{D}}_{\text{ray},\Gamma}$ is invariant under a constant parabolic rescaling of the represented window. Equivalently, it is defined by the Radon–Nikodym relation

$$(8.29) \quad \widehat{\mathfrak{D}}_{\text{ray},\Gamma}(\tau) \, d\tau := \Lambda(\tau)^{-1} \, \text{dist}_{K^{-1}}^2(J_\nu, \mathbb{R}_+ G_\Gamma) \, dt.$$

Proof. Both vectors in the Stokes pairing transform by the pullback of the same chart. Their squared inverse-metric distance has the scaling of viscous action. Formula (5.11) shows that its physical measure is Λ times the dimensionless represented measure, which is exactly (8.29). $\square$

The hatted density exhibits the same physical-versus-critical distinction as (6.3) and (6.4). It is retained below as a covariant diagnostic, but the local coercive floor does not use it.

9. CANONICAL CRITICAL ORBITS

The covariant argument is carried by one normalized orbit rather than a list of asymptotic regimes. This section specifies the state space and the exact compactness output required by the entropy proof.

9.1. Symmetry reduction and bad windows. Fix a normalized probe radius $R_* > 1$ and a nonnegative inner cutoff $\zeta \in C_c^\infty(B_{R_*})$, $0 \leq \zeta \leq 1$, $\int \zeta^3 > 0$, for which $\{\zeta > 0\}$ is connected and whose support lies where the adjoint cutoff equals one. Put

$$(9.1) \quad \bar{V}_\zeta(\tau) := \frac{\int \zeta^3 V(y, \tau) \, dy}{\int \zeta^3 \, dy}.$$

For a unit represented interval $I = [s, s+1]$, define

$$(9.2) \quad \mathcal{C}_\Gamma^\circ(V; I) := \int_I \int_{B_{R_*}} \zeta^3 |V - \bar{V}_\zeta|^3 \, dy \, d\tau.$$

This observable is independent of the adjoint terminal datum, continuous under strong local L^3 convergence, and invariant under constant velocity shifts.

The relevant symmetries are spatial translation, positive parabolic dilation, constant-velocity Galilean transformation, and addition of a function of time to the pressure. The repaired variables (X, Λ, a, b) encode scale and center. We fix those representatives by

$$(9.3) \quad \mathfrak{F}(1, 0; V(\tau)) = 0, \quad \det D_{(\lambda, q)} \mathfrak{F}(1, 0; V(\tau)) \neq 0$$

at almost every active time, with absolutely continuous continuation supplied by the orbit bridge. Galilean symmetry is not fixed by (9.3); the gauge itself is invariant under it, and the scalar observables remove it through (9.1) and (8.3). The pressure gauge is

$$(9.4) \quad \int_{B_1} P(y, \tau) \, dy = 0.$$

The nondegeneracy and absolute continuity of this gauge are conclusions of the critical-orbit bridge, not extra hypotheses imposed on an arbitrary weak solution.

Definition 9.1 (Normalized bad window). Fix $\eta_0 > 0$ and a collection of compact-cylinder bounds $\mathbf{M} = (M_R)_{R \geq 1}$. A represented unit window $\mathfrak{w} = (V, P, a, b; I)$ is an $(\eta_0, \mathbf{M})$ -normalized bad window when:

- (i) (V, P) is a suitable solution of (5.4) on a neighborhood of $B_R \times I$ for every fixed R ;
- (ii) the gauges (9.3) and (9.4) hold, and the inverse of the gauge Jacobian in (6.9) is uniformly bounded on the window;
- (iii) the local suitable norms satisfy, on each fixed $B_R \times I$,

$$(9.5) \quad \|V\|_{L_I^\infty L^2(B_R)} + \|V\|_{L_I^2 H^1(B_R)} + \|V\|_{L^3(B_R \times I)} + \|P\|_{L^{3/2}(B_R \times I)} + \|a\|_{L^1(I)} + \|b\|_{L^1(I)} \leq M_R;$$

(iv) the oscillatory floor

$$(9.6) \quad \mathcal{C}_T^\circ(V; I) \geq \eta_0$$

holds for the selected inner probe.

The subtraction in (9.2) is essential. A spatially constant represented velocity can have positive uncentered L^3 mass, but it has no spatial oscillation and cannot by itself support a singular point; constant Galilean shifts are a special case. The independent orbit construction must therefore prove that singular concentration survives the variance reduction rather than assume this property.

9.2. The critical-orbit bridge.

Bridge hypothesis 9.2 (Canonical critical orbit from terminal loss). *Let (u, p) be a global finite-energy suitable weak solution of (1.1), and let $0 < T < \infty$. Suppose either*

- (a) $x_* \in \Sigma_u(T)$ for some finite point x_* , or
- (b) u is the suitable continuation of a smooth finite-energy solution whose maximal classical existence time is T .

Then there exist $\eta_0 = \eta_0(\nu) > 0$, compact bounds $\mathbf{M}$, an absolutely continuous repaired chart (X, Λ) , a represented suitable orbit

$$(V, P, a, b)(\tau), \quad \tau \in [0, \infty),$$

unit intervals $I_j = [\tau_j, \tau_j + 1]$ with pairwise disjoint interiors and $\tau_j \rightarrow \infty$, with the following properties:

- (i) *after passing once to a subsequence and relabeling, with $x_j := X(\tau_j)$ and $r_j := \Lambda(\tau_j)$, one has $r_j \downarrow 0$; the physical time images of the windows converge to T ; in case (a) one may take $x_j \rightarrow x_*$, while case (b) also permits $|x_j| \rightarrow \infty$;*
- (ii) *every restricted state $\mathfrak{w}_j = (V, P, a, b; I_j)$ is an $(\eta_0, \mathbf{M})$ -normalized bad window;*
- (iii) *in the product topology specified in Section 6 and made explicit here, the sequential closure of the selected windows is sequentially compact. More explicitly, after translating each time interval to $[0, 1]$, every sequence in that closure has a subsequence for which*

$$(9.7) \quad V_j \rightarrow V_\infty \text{ strongly in } L_{\text{loc}}^3, \quad \nabla V_j \rightharpoonup \nabla V_\infty \text{ in } L_{\text{loc}}^2,$$

on every compact spatial cylinder in $\mathbb{R}^3 \times [0, 1]$, and in particular strongly in $L^3(\text{supp } \zeta \times [0, 1])$, with strong local $L^{3/2}$ convergence of normalized pressures and L^1 convergence of (a, b) ;

- (iv) *chart changes obey a cocycle law and represent the same physical solution.*

PHASE-I SOURCE. The trilogy supplies precursor results on velocity concentration, profile extraction, retained cores, repaired gauges, pressure, and compactness. It does not state the single coherent orbit above. Accordingly, the displayed statement is a bridge hypothesis rather than an imported theorem.

INDEPENDENT PROOF CONTRACT. Appendix C requires construction of the coherent orbit from the precursor results, and Appendix D requires construction of the gauge, pressure atlas, and moving carrier. The replacement must establish all four clauses without using a final exclusion theorem.

Remark 9.3 (Why one orbit is required). The carrier loss is accumulated along a scalar function of represented time. Independent rescaling on each bad window would provide unrelated endpoint values and would not support the telescoping argument. Clause (iv) is therefore part of the orbit bridge rather than a convention made after extraction.

9.3. The two classical scale regimes. Let $\ell(t) = \sqrt{T-t}$ denote the parabolic terminal scale. Along the canonical orbit, the dimensionless ratio

$$(9.8) \quad \mathfrak{r}(\tau) := \frac{\Lambda(\tau)}{\ell(t(\tau))}$$

records the classical distinction. A Type I window has $\mathfrak{r}$ bounded above and below after normalization; in the centered choice $t = -e^{-\tau}$, $\Lambda = e^{-\tau/2}$, one has $a = -1/2$. A Type II sequence contains windows on which $\mathfrak{r} \rightarrow 0$, and its coefficient $a = \Lambda'/\Lambda$ must be controlled through the repaired gauge (6.8).

The covariant equation (5.6) is identical in the two regimes. The difference is carried entirely by (a, b) and by the asymptotic path of the normalized orbit. The carrier alternative is consequently formulated for locally integrable a, b , without fixing $a = -1/2$ or assuming convergence of a . A centered ancient orbit, a rapid scale collapse, and a recurrent repaired orbit are either routed to their named local closures or tested by the same moving production measure. The gap argument uses only the coherent orbit, compactness, the oscillatory floor, and the summable carrier branch. Thus the second-generation argument takes compactness as the exact bridge output and introduces no additional descendant object.

There remains one compatibility issue. A window selected at time τ_j must extend to the next selected window without replacing the underlying solution. Let $g_{j+1 \leftarrow j} \in \mathcal{G}_{\text{ch}}$ be the accumulated canonical transition from the terminal representative of I_j to the initial representative of I_{j+1} . Coherence means

$$(9.9) \quad g_{k \leftarrow j} = g_{k \leftarrow k-1} \circ \cdots \circ g_{j+1 \leftarrow j}, \quad j < k,$$

and every composition pulls back the same physical pair (u, p) . The moving cutoff in [Bridge hypothesis 11.5](#) is defined on this one orbit, so its corrected energy is one global scalar. There is no cross-window adjoint reset in the exclusion argument.

10. THE COVARIANT NO-LOSS CASCADE PRINCIPLE

The material identities identify the mechanism but do not by themselves produce a scale-critical finite measure. The required statement is a no-loss theorem: signed transport between nested regions is internal, whereas viscous expenditure and true exterior leakage are charged once. Its precise interface with the compact orbit is derived below from the localized energy equation.

Let $I = [s, t]$ be a represented interval and let $\psi \in C^\infty(I; C_c^\infty(\mathbb{R}^3))$, $\psi \geq 0$. Set

$$(10.1) \quad K_\psi(\tau) := \frac{1}{2} \int_{\mathbb{R}^3} |V(y, \tau)|^2 \psi(y, \tau) dy, \quad D_I(\psi) := \nu \int_I \int_{\mathbb{R}^3} |\nabla V|^2 \psi dy d\tau,$$

and define the signed localization current

$$(10.2) \quad J_I(\psi) := \int_I \left[\frac{1}{2} \int |V|^2 (\partial_\tau \psi + \nu \Delta \psi) + \int \left(\frac{1}{2} |V|^2 + P \right) V \cdot \nabla \psi - \frac{a}{2} \int |V|^2 (\psi + y \cdot \nabla \psi) - \frac{1}{2} b \cdot \int |V|^2 \nabla \psi \right] d\tau.$$

All spatial integrals in (10.2) are over $\mathbb{R}^3$. The pressure contribution is invariant under $P \mapsto P + c(\tau)$, because $\int V \cdot \nabla \psi = 0$.

Proposition 10.1 (Signed-current additivity). *For every represented suitable solution there is a nonnegative local energy-defect functional $M_I(\psi)$, linear on nonnegative tests, such that*

$$(10.3) \quad K_\psi(t) + D_I(\psi) + M_I(\psi) \leq K_\psi(s) + J_I(\psi).$$

For a smooth solution equality holds and $M_I(\psi) = 0$. If $\psi = \sum_{k=1}^N \psi_k$ with $\psi_k \geq 0$, then

$$(10.4) \quad K_\psi = \sum_{k=1}^N K_{\psi_k}, \quad D_I(\psi) = \sum_{k=1}^N D_I(\psi_k), \quad J_I(\psi) = \sum_{k=1}^N J_I(\psi_k), \quad M_I(\psi) = \sum_{k=1}^N M_I(\psi_k).$$

Consequently, for an exact parent–child partition $\psi_\alpha = \sum_{\beta \in \text{ch}(\alpha)} \psi_\beta$, the difference between the parent current and the sum of its child currents is identically zero. In particular, pressure and convection cannot be charged once at the parent scale and again at the child scales.

Proof. The suitable local energy inequality for the repaired equation, integrated between s and t , is precisely (10.3); the terms generated by the repaired drift are the last two terms of (10.2). The nonnegative distribution by which equality fails defines M_I . Every displayed term is linear in ψ and its derivatives, which proves (10.4). For smooth solutions the local energy identity holds with equality. $\square$

Definition 10.2 (Navier–Stokes covariant expenditure). A covariant expenditure on a coherent orbit is a nonnegative Radon measure obtained by summing $D_I + M_I$ over a compatible nested family of tests after the internal currents in (10.4) have been canceled. Only the root current, terminal leaf energy, and flux through the true exterior boundary may remain in its global upper bound. A window is *lossless* when the lower-semicontinuous local cost dominated by this measure vanishes.

[Proposition 10.1](#) proves the finite-tree algebra. It does not assert the existence of an infinite compatible tree, finiteness of its exterior input, or rigidity of its lossless windows. Those are the Navier–Stokes analytic assertions collected in the following bridge and developed independently in [Appendices F](#) and [G](#).

Bridge hypothesis 10.3 (Covariant no-loss cascade). *For every terminal-loss orbit supplied by [Bridge hypothesis 9.2](#), one of the following alternatives holds.*

- (N1) *The orbit enters a certified material reduction or a named local closure that is incompatible with a terminal singular branch.*
- (N2) *There exist a finite nonnegative Radon measure $\mathfrak{m}_{\text{mathrm{cov}}}$ on a terminal tail $[\tau_0, \infty)$ and a sequentially lower semicontinuous functional*

$$(10.5) \quad \mathfrak{d}_{\text{cov}} : \overline{\{\mathfrak{w}_j : \tau_j \geq \tau_0\}} \longrightarrow [0, \infty]$$

such that

$$(10.6) \quad \mathfrak{m}_{\text{mathrm{cov}}}([\tau_0, \infty)) < \infty,$$

$$(10.7) \quad \mathfrak{m}_{\text{mathrm{cov}}}(I_j) \geq \mathfrak{d}_{\text{cov}}(\mathfrak{w}_j),$$

$$(10.8) \quad \mathfrak{d}_{\text{cov}}(\mathfrak{w}) = 0 \implies \mathfrak{w} \text{ is oscillation-removable and } \mathcal{C}_\Gamma^\circ(V; [0, 1]) = 0.$$

Here the closure and its topology are those of [Bridge hypothesis 9.2](#), after translating each unit interval to $[0, 1]$.

The content of the two alternatives is rigidity of lossless concentration transfer. A zero-cost limit is either routed to the certified reduced closure (N1) or, on the surviving paid branch, satisfies (10.8). Thus an orbit cannot retain critical oscillation while passing its activity to smaller scales with zero covariant expenditure. No numerical coercive inequality is postulated. Once the equality classes are identified, compactness supplies the numerical gap.

Theorem 10.4 (No-loss window gap). *Assume alternative (N2) and let $\eta_0 > 0$ be the oscillation floor of the canonical orbit. There exists $\delta_{\text{cov}} > 0$ such that every selected window in the terminal tail satisfies*

$$(10.9) \quad \mathfrak{m}_{\text{mathrm{cov}}}(I_j) \geq \delta_{\text{cov}}.$$

Proof. The compact closure of the selected windows contains the closed subset

$$\mathcal{B}_{\eta_0} := \{\mathbf{w} : \mathcal{C}_\Gamma^\circ(V; [0, 1]) \geq \eta_0\}.$$

It is closed because the velocities converge strongly in L^3 on the fixed retained core. If the infimum of $\mathfrak{d}_{\text{cov}}$ on $\mathcal{B}_{\eta_0}$ were zero, compactness and lower semicontinuity would produce $\mathbf{w}_\infty \in \mathcal{B}_{\eta_0}$ with $\mathfrak{d}_{\text{cov}}(\mathbf{w}_\infty) = 0$. This contradicts (10.8). Hence the infimum is a positive number δ_{cov} , and (10.7) proves (10.9). $\square$

Remark 10.5 (Phase-I and independent realizations). The phase-I realization takes $\mathfrak{m}_{\text{mathrm{cov}}} = \mathfrak{D}_{\text{NS}}$ from the corrected moving carrier and $\mathfrak{d}_{\text{cov}} = \frac{1}{2}\mathfrak{d}_\nu$, where $\mathfrak{d}_\nu$ is the fixed inner Dirichlet cost (12.3). The trilogy supplies the flux routing needed to make this one finite measure. The independent realization sought in Appendices F–G keeps convection, pressure, and gauge motion signed until neighboring scale-space cells are summed. Internal currents then cancel, and (10.8) becomes the classification of the remaining lossless material geometry.

11. COVARIANT ADJOINT GEOMETRY AND THE MOVING CARRIER

The adjoint entropy $\mathcal{W}_{\text{ad}}$, the kinetic variance $\mathcal{E}_\Gamma^\circ$, and the ray defect $\mathcal{D}^{\text{ray}}$ have distinct roles. The first controls the localization geometry, the second detects the retained velocity core, and the third measures the component of viscous motion transverse to the target-gradient ray. None of them separately has the monotonicity and coercivity needed below.

11.1. The explicit nonnegative components. For a smooth positive adjoint density, define the Gaussian production

$$(11.1) \quad \mathfrak{D}_G(\rho) := 2\nu^2\theta \int_{B_R} \chi^2 \left| \nabla^2 f - \frac{1}{2\nu\theta} I \right|^2 \rho \, dy.$$

Define also the localized viscous production

$$(11.2) \quad \mathfrak{D}_{\nu,\Gamma}(V) := \nu \int_{B_R} m |\nabla V|^2 \, dy$$

and the instantaneous ray production

$$(11.3) \quad \mathfrak{D}_{\text{ray},\Gamma}(V) := \widehat{\mathfrak{D}}_{\text{ray},\Gamma}(V),$$

where the hatted, scale-critical density is defined by (8.29). At weak regularity $\mathfrak{D}_{\nu,\Gamma}$ is defined directly from the weighted Dirichlet integral. If the Gaussian or ray terms are retained in a weak production measure, their identification and compact stability require additional adjoint compactness beyond the carrier argument; neither is inferred merely from local weak velocity convergence. The coercive floor used in the compactness argument is independent of the adjoint density:

$$(11.4) \quad \mathfrak{D}_{0,\zeta}(V) := \nu \int_{\mathbb{R}^3} \zeta^2 |\nabla V|^2 \, dy.$$

The Gaussian production remains a favorable nonnegative term in the smooth Perelman identity, but it may be discarded before taking a weak limit. The ray production is likewise an additional nonnegative covariant diagnostic on each finite-energy representative. Neither term is needed to separate retained compact limits from equality. Inserting the ray term into (11.4) would in addition require a localization theorem for the whole-space Helmholtz projection under merely local blow-up convergence.

Formula (7.17) contains a raw velocity-strain term with no separate sign. It becomes coercive after coupling it to the viscous and Gaussian terms in Proposition 11.2. The moving kinetic

carrier below handles pressure and cutoff transport directly and dominates (11.4) because $\Phi = 1$ on $\text{supp } \zeta$.

11.2. The uncoupled entropy ledger. The whole-space smooth calculation identifies every term that the correction must address. Put

$$(11.5) \quad \mathcal{L}_c F := \partial_\tau F - \nu \Delta F + c \cdot \nabla F, \quad c = V - ay - b.$$

For the kinetic density $e = |V|^2/2$, the repaired equation gives

$$(11.6) \quad \mathcal{L}_c e = -\nu |\nabla V|^2 + 2ae - \text{div}(PV).$$

Indeed, after moving the dilation and translation derivatives to the left, (5.4) becomes

$$\partial_\tau V + c \cdot \nabla V + \nabla P = \nu \Delta V + aV.$$

Taking its scalar product with V proves (11.6).

Proposition 11.1 (Exact uncoupled entropy ledger). *Let (V, P) and a positive probability density ρ be smooth on $\mathbb{R}^3$, with sufficient decay, and suppose that ρ solves the unnormalized whole-space adjoint equation. Set*

$$\mathcal{K}_\rho(\tau) := \frac{1}{2} \int_{\mathbb{R}^3} |V|^2 \rho \, dy.$$

Then

$$(11.7) \quad \frac{d}{d\tau} \mathcal{K}_\rho = -\nu \int |\nabla V|^2 \rho + 2a\mathcal{K}_\rho - \int PV \cdot \nabla f \, \rho.$$

For every $\beta > 0$, the uncoupled combination $\mathcal{H}_\beta := \mathcal{K}_\rho - \beta \mathcal{W}_{\text{ad}}$ satisfies

$$(11.8) \quad \begin{aligned} \frac{d}{d\tau} \mathcal{H}_\beta = & -\nu \int |\nabla V|^2 \rho - 2\beta \nu^2 \theta \int \left| \nabla^2 f - \frac{1}{2\nu\theta} I \right|^2 \rho \\ & - \int PV \cdot \nabla f \, \rho + 2a\mathcal{K}_\rho \\ & + 2\beta \nu \theta \int (\nabla f)^T (\nabla V)_{\text{sym}} \nabla f \, \rho. \end{aligned}$$

Proof. Apply the whole-space form of Lemma 7.3 to (11.6). Integration by parts gives

$$- \int \text{div}(PV) \rho = \int PV \cdot \nabla \rho = - \int PV \cdot \nabla f \, \rho,$$

which proves (11.7). Subtract β times (7.17) and collect terms. $\square$

The strain term in (11.8) combines exactly with the viscous and Gaussian terms. Because the heat scale obeys (7.7), no scalar modulation term remains in the adjoint part of the identity.

Proposition 11.2 (Strain–Gaussian square completion). *Under the hypotheses of Proposition 11.1, set*

$$S := \frac{1}{2} (\nabla V + (\nabla V)^T), \quad \Omega := \frac{1}{2} (\nabla V - (\nabla V)^T), \quad A_f := \nabla^2 f - \frac{1}{2\nu\theta} I.$$

Then

$$(11.9) \quad \int_{\mathbb{R}^3} \rho S : (\nabla f \otimes \nabla f) \, dy = \int_{\mathbb{R}^3} \rho S : A_f \, dy.$$

Consequently the entropy ledger has the exact form

$$(11.10) \quad \frac{d}{d\tau} \mathcal{H}_\beta + \mathfrak{Q}_{\beta, \theta}(V, f) = - \int_{\mathbb{R}^3} PV \cdot \nabla f \, \rho \, dy + 2a\mathcal{K}_\rho,$$

where

$$(11.11) \quad \begin{aligned} \mathfrak{Q}_{\beta,\theta}(V, f) := & \nu \int_{\mathbb{R}^3} |\Omega|^2 \rho \, dy + \left(\nu - \frac{\beta\theta}{2} \right) \int_{\mathbb{R}^3} |S|^2 \rho \, dy \\ & + 2\beta\nu^2\theta \int_{\mathbb{R}^3} \left| A_f - \frac{1}{2\nu} S \right|^2 \rho \, dy. \end{aligned}$$

If $0 < \beta\theta \leq \nu$, then

$$(11.12) \quad \mathfrak{Q}_{\beta,\theta}(V, f) \geq \frac{\nu}{2} \int_{\mathbb{R}^3} |\nabla V|^2 \rho \, dy.$$

In particular, the fixed choice

$$(11.13) \quad 0 < \beta \leq \frac{\nu}{(1 + \kappa_*)e^{2M_*}}$$

ensures (11.12) throughout every retained window satisfying (7.6).

Proof. Since $\rho = (4\pi\nu\theta)^{-3/2}e^{-f}$,

$$(11.14) \quad \nabla^2 \rho = \rho(\nabla f \otimes \nabla f - \nabla^2 f).$$

Incompressibility gives

$$\partial_i \partial_j S_{ij} = \Delta \operatorname{div} V = 0.$$

The decay hypotheses therefore justify two integrations by parts and give

$$\int_{\mathbb{R}^3} S : \nabla^2 \rho \, dy = \int_{\mathbb{R}^3} \rho \partial_i \partial_j S_{ij} \, dy = 0.$$

Using (11.14) and $\operatorname{tr} S = \operatorname{div} V = 0$ yields

$$\int \rho S : (\nabla f \otimes \nabla f) = \int \rho S : \nabla^2 f = \int \rho S : A_f.$$

This proves (11.9). Substitution into (11.8) leaves the right-hand side of (11.10). Finally,

$$\begin{aligned} & \nu |\nabla V|^2 + 2\beta\nu^2\theta |A_f|^2 - 2\beta\nu\theta S : A_f \\ & = \nu |\Omega|^2 + \left(\nu - \frac{\beta\theta}{2} \right) |S|^2 + 2\beta\nu^2\theta \left| A_f - \frac{S}{2\nu} \right|^2, \end{aligned}$$

because $|\nabla V|^2 = |S|^2 + |\Omega|^2$. This proves (11.11). When $\beta\theta \leq \nu$, the first two terms on its right control $(\nu/2)(|S|^2 + |\Omega|^2)$, proving (11.12). The last assertion follows from the upper bound for θ in (7.6). $\square$

Lemma 11.3 (Localized strain transmutation). *Let V, ρ, f be smooth on B_R , with $\operatorname{div} V = 0$, let $\chi \in C_c^\infty(B_R)$, and assume $\rho = (4\pi\nu\theta)^{-3/2}e^{-f}$ and $\int_{B_R} \chi^2 \rho = 1$. With A_f as in Proposition 11.2,*

$$(11.15) \quad \int_{B_R} \chi^2 \rho S : (\nabla f \otimes \nabla f) \, dy = \int_{B_R} \chi^2 \rho S : A_f \, dy + \mathfrak{C}_\chi(S, \rho),$$

$$(11.16) \quad \mathfrak{C}_\chi(S, \rho) := - \int_{B_R} \rho S : \nabla^2(\chi^2) \, dy - 2 \int_{B_R} S : (\nabla(\chi^2) \otimes \nabla \rho) \, dy.$$

Writing $\mathcal{A}_\chi := \text{supp}(|\nabla\chi| + |\nabla^2\chi|)$, one has for every $\hat{\chi} \in C_c^\infty(B_R)$ satisfying $\hat{\chi} = 1$ on a neighborhood of $\mathcal{A}_\chi$,

$$(11.17) \quad \begin{aligned} |\mathfrak{C}_\chi(S, \rho)| &\leq \|\nabla^2(\chi^2)\|_\infty \int_{\mathcal{A}_\chi} \rho |S| \, dy \\ &\quad + 2\|\nabla(\chi^2)\|_\infty \left(\int_{\mathcal{A}_\chi} \rho |S|^2 \, dy \right)^{1/2} \left(\int_{B_R} \hat{\chi}^2 \frac{|\nabla\rho|^2}{\rho} \, dy \right)^{1/2} \end{aligned}$$

Moreover, with $m = \chi^2 \rho$, the localized quadratic form

$$(11.18) \quad \begin{aligned} \mathfrak{Q}_{\beta, \theta}^\chi(V, f) &:= \nu \int_{B_R} m |\Omega|^2 \, dy + \left(\nu - \frac{\beta\theta}{2} \right) \int_{B_R} m |S|^2 \, dy \\ &\quad + 2\beta\nu^2\theta \int_{B_R} m \left| A_f - \frac{S}{2\nu} \right|^2 \, dy \end{aligned}$$

satisfies, whenever $0 < \beta\theta \leq \nu$,

$$(11.19) \quad \mathfrak{Q}_{\beta, \theta}^\chi(V, f) \geq \frac{\nu}{2} \int_{B_R} m |\nabla V|^2 \, dy = \frac{1}{2} \mathfrak{D}_{\nu, \Gamma}(V).$$

Proof. Identity (11.14) holds on B_R . Moreover $\partial_i \partial_j S_{ij} = 0$, as in the proof of Proposition 11.2. Testing this distributional identity against the compactly supported function $\chi^2 \rho$ gives

$$0 = \int_{B_R} S : \nabla^2(\chi^2 \rho) \, dy.$$

Expanding the Hessian and using the symmetry of S yields

$$\int_{B_R} \chi^2 S : \nabla^2 \rho \, dy = - \int_{B_R} \rho S : \nabla^2(\chi^2) \, dy - 2 \int_{B_R} S : (\nabla(\chi^2) \otimes \nabla \rho) \, dy.$$

Substitute (11.14), then use $\text{tr } S = 0$. This proves (11.15) and (11.16). The first term in (11.17) follows directly from support and the L^∞ norms of the cutoff derivatives. For the second, use

$$|S| |\nabla \rho| = (\sqrt{\rho} |S|) \left(\frac{|\nabla \rho|}{\sqrt{\rho}} \right)$$

and Cauchy–Schwarz on $\mathcal{A}_\chi$. The last integral over $\mathcal{A}_\chi$ is bounded by the displayed $\hat{\chi}$ -weighted Fisher information because $\hat{\chi} = 1$ there. The identity $|\nabla V|^2 = |S|^2 + |\Omega|^2$ and the same pointwise square completion as in Proposition 11.2 prove (11.18)–(11.19). $\square$

Thus velocity strain requires no independent correction in the interior smooth identity. The only localized discrepancy is the annular commutator $\mathfrak{C}_\chi$. The pressure pairing and the remaining kinetic scale term in (11.10), however, have no sign. Rather than force them into an adjoint primitive, we now use the local kinetic-energy identity, where pressure and modulation already occur as cutoff fluxes. This is the decisive separation: the Perelman functional supplies covariant geometry, while the moving carrier supplies monotonicity at suitable-weak regularity.

In particular, the adjoint pairing $-\int PV \cdot \nabla f \rho$ is not identified with the moving-cutoff flux $\int PV \cdot \nabla \Phi$. The pressure-tail estimates apply to the latter. The proof therefore discards the adjoint ledger before invoking monotonicity; no cancellation between these two distinct pressure terms is claimed.

11.3. The moving kinetic carrier. Fix a radial nonincreasing $\phi \in C_c^\infty(B_2)$, with $0 \leq \phi \leq 1$ and $\phi = 1$ on B_1 . Let $R : [\tau_0, \infty) \rightarrow [R_*, \infty)$ be locally absolutely continuous and nondecreasing, and put

$$(11.20) \quad \Phi(\tau, y) := \phi(y/R(\tau)), \quad \mathcal{E}_R(\tau) := \frac{1}{2} \int_{\mathbb{R}^3} |V|^2 \Phi \, dy.$$

Write $a_+ = \max\{a, 0\}$, $a_- = \max\{-a, 0\}$, and define

$$(11.21) \quad \tilde{\mathfrak{D}}_R(\tau) := \nu \int |\nabla V|^2 \Phi \, dy + a_+(\tau) \int |V|^2 \Phi \, dy.$$

The full unsigned boundary-flux density is

$$(11.22) \quad \begin{aligned} B_{\text{mov}}(\tau) := & \frac{1}{2} \left| \int |V|^2 \partial_\tau \Phi \right| + \frac{\nu}{2} \left| \int |V|^2 \Delta \Phi \right| + \frac{1}{2} \left| \int |V|^2 V \cdot \nabla \Phi \right| + \left| \int P V \cdot \nabla \Phi \right| \\ & + \frac{1}{2} |b(\tau)| \left| \int |V|^2 \nabla \Phi \right| + \frac{1}{2} a_-(\tau) \int |V|^2 \Phi + \frac{1}{2} a_+(\tau) \int |V|^2 (-y \cdot \nabla \Phi). \end{aligned}$$

Every integral in (11.21)–(11.22) is over $\mathbb{R}^3$. The pressure term is unchanged by adding a function of time to P , because $\int V \cdot \nabla \Phi = 0$.

Theorem 11.4 (Corrected moving-carrier monotonicity). *Let (V, P) be a represented suitable weak solution of (5.4). If $B_{\text{mov}} \in L^1(\tau_0, \infty)$, then*

$$(11.23) \quad \mu_{\text{NS}}(\tau) := \mathcal{E}_R(\tau) + \int_\tau^\infty B_{\text{mov}}(s) \, ds$$

has a nonincreasing representative and, in distributions,

$$(11.24) \quad \frac{d}{d\tau} \mu_{\text{NS}} + \frac{1}{2} \tilde{\mathfrak{D}}_R \leq 0.$$

In particular $\mu_{\text{NS}} \geq 0$, and for $\tau_0 \leq s < t$,

$$(11.25) \quad \mu_{\text{NS}}(t) + \mathfrak{D}_{\text{NS}}((s, t]) \leq \mu_{\text{NS}}(s), \quad \mathfrak{D}_{\text{NS}}((s, t]) := \frac{1}{2} \int_s^t \tilde{\mathfrak{D}}_R(\tau) \, d\tau.$$

Proof. Testing the suitable local energy inequality with Φ gives

$$\begin{aligned} \frac{d}{d\tau} \mathcal{E}_R \leq & -\nu \int |\nabla V|^2 \Phi + \frac{1}{2} \int |V|^2 \partial_\tau \Phi + \frac{\nu}{2} \int |V|^2 \Delta \Phi + \frac{1}{2} \int |V|^2 V \cdot \nabla \Phi \\ & + \int P V \cdot \nabla \Phi - \frac{a}{2} \int |V|^2 \Phi - \frac{a}{2} \int |V|^2 y \cdot \nabla \Phi - \frac{1}{2} b \cdot \int |V|^2 \nabla \Phi. \end{aligned}$$

Add $\tilde{\mathfrak{D}}_R/2$. The viscous term leaves $-\frac{\nu}{2} \int |\nabla V|^2 \Phi \leq 0$, while $-aM/2 + a_+M/2 = a_-M/2$. Since ϕ is radial nonincreasing, $y \cdot \nabla \Phi \leq 0$; hence the scale-shell term is bounded above by the last term of (11.22). Taking absolute values of the other fluxes proves

$$\frac{d}{d\tau} \mathcal{E}_R + \frac{1}{2} \tilde{\mathfrak{D}}_R \leq B_{\text{mov}}.$$

The suitable energy-defect measure has the favorable sign and was discarded in this inequality. Differentiating the finite tail in (11.23) subtracts B_{mov} , proving (11.24). Both terms in (11.23) are nonnegative. $\square$

11.4. Canonical moving-carrier bridge.

Bridge hypothesis 11.5 (Canonical moving-carrier alternative). *For every putative terminal-loss orbit supplied by [Bridge hypothesis 9.2](#), one of the following two alternatives holds.*

- (C1) *The orbit enters a named local alternative already incompatible with a terminal singular branch: pressure reconstruction or exterior leakage, multicore or same-point cascade, compact scale collapse, scale rigidity, cost divergence, thin negative drift, or the terminal Type I alternative.*
- (C2) *After restriction to a terminal tail $[\tau_0, \infty)$, there is a single nondecreasing locally absolutely continuous radius $R(\tau) \geq R_*$, with $R(\tau) \rightarrow \infty$, such that the cutoff Φ in [\(11.20\)](#) satisfies*

$$(11.26) \quad B_{\text{mov}} \in L^1(\tau_0, \infty).$$

The radius is chosen so that $\Phi = 1$ on B_{R_} , and hence on $\text{supp } \zeta$, for all $\tau \geq \tau_0$.*

Remark 11.6 (Construction required by the hypothesis). The moving monotonicity theorem itself is proved above. The bridge contains only the global scheduling and routing statement. In the companion trilogy, the pressure-tail lemma, tailored radius schedule, and moving-family estimate control the diffusion-shell, convection, pressure, and translation terms; the transition, positive-scale, cost-divergence, and negative-drift theorems route the remaining nonsummable components. Appendix E records the independent proof contract for precisely this synthesis. No adjoint reset law and no pressure correction to $\mathcal{W}_{\text{ad}}$ is required.

Proposition 11.7 (Phase-I discharge of the carrier alternative). *Assume the exhaustive local branch partition and the results cited in the trilogy source ledger. Then every canonical orbit that reaches the retained moving-carrier row satisfies [Bridge hypothesis 11.5](#).*

Proof. The canonical pressure decomposition and modulation system give the repaired local energy identity with a pressure representative stable under the moving tests. Lemmas 8.114–8.116 of [4] choose the tailored radius and make the diffusion-shell, convection, pressure, and translation components of [\(11.22\)](#) summable. Thus, by Corollary 8.117 there, only the transition component, the positive scale-shell component, and the negative scale-core component remain.

If the transition or positive scale-shell component is not integrable on any tail, Corollaries 8.126 and 8.131 of [4] route the orbit to a named exterior, recentering, or scale branch. For the negative scale-core component, finite cost is precisely integrability; divergent cost is closed by Theorem 10.42, while the diffuse residual alternative is removed by Theorem 10.118 of [4]. The exhaustive branch partition closes the other named exits, including the terminal Type I branch. These are alternative (C1). If none occurs, all seven nonnegative summands of [\(11.22\)](#) are integrable on one common later tail, which is alternative (C2). $\square$

11.5. Equality states.

Definition 11.8 (Oscillation-removable equality state). A represented state is oscillation-removable when its velocity is spatially constant on the connected retained region $\{\zeta > 0\}$ for almost every represented time. The constant may depend on time. Such a state has zero Galilean-reduced oscillatory concentration. No assertion that a localized Dirichlet adjoint density equals the whole-space Gaussian is included in this definition; the Gaussian remains the equality model only for the whole-space identity [\(7.16\)](#).

Theorem 11.9 (Inner-Dirichlet and zero-production rigidity). *Let $\mathfrak{w} = (V, P, a, b; [0, 1])$ be a normalized window or a compact limit of normalized windows. If*

$$(11.27) \quad \int_0^1 \mathfrak{D}_{0,\zeta}(V(\tau)) \, d\tau = 0,$$

then $\mathfrak{w}$ is oscillation-removable and

$$(11.28) \quad \mathcal{C}_\Gamma^\circ(V; [0, 1]) = 0.$$

In particular, zero moving-carrier production on the window implies (11.28) whenever $\Phi = 1$ on $\text{supp } \zeta$.

Proof. Since $\nu > 0$, (11.27) gives $\nabla V = 0$ almost everywhere on the connected set $\{\zeta > 0\}$ for almost every τ . Hence $V(\cdot, \tau) = c(\tau)$ there. Its ζ^3 -weighted mean is the same vector $c(\tau)$, so the integrand in (9.2) vanishes. This proves (11.28). The final assertion follows from the bound $\mathfrak{D}_{\text{NS}}(I) \geq \frac{1}{2} \int_I \mathfrak{D}_{0, \zeta}$ and nonnegativity. $\square$

Proposition 11.10 (Phase-I discharge of the no-loss bridge). *Assume the canonical moving-carrier alternative [Bridge hypothesis 11.5](#) and the trilogy closures named in alternative (C1). Then the covariant no-loss cascade bridge [Bridge hypothesis 10.3](#) holds.*

Proof. Alternative (C1) is alternative (N1). In alternative (C2), let

$$\mathfrak{m}_{\text{mathrm{mcov}}}((s, t]) := \mathfrak{D}_{\text{NS}}((s, t]) = \frac{1}{2} \int_s^t \tilde{\mathfrak{D}}_R(\tau) \, d\tau$$

on the terminal tail. The carrier ledger (11.25) and $\mu_{\text{NS}} \geq 0$ give

$$\mathfrak{m}_{\text{mathrm{mcov}}}([\tau_0, \infty)) \leq \mu_{\text{NS}}(\tau_0) < \infty.$$

For a unit window translated to $[0, 1]$, set

$$\mathfrak{d}_{\text{cov}}(\mathfrak{w}) := \frac{1}{2} \mathfrak{d}_\nu(\mathfrak{w}) = \frac{\nu}{2} \int_0^1 \int \zeta^2 |\nabla V|^2 \, dy \, d\tau.$$

Because $\Phi = 1$ on $\text{supp } \zeta$, one has $\mathfrak{m}_{\text{mathrm{mcov}}}(I_j) \geq \mathfrak{d}_{\text{cov}}(\mathfrak{w}_j)$. Lower semicontinuity is (A.8). If $\mathfrak{d}_{\text{cov}} = 0$, [Theorem 11.9](#) makes the state spatially constant on the retained connected core and hence oscillation-removable. These are exactly (10.6)–(10.8). $\square$

12. PHASE-I MOVING-CARRIER REALIZATION OF THE GAP

This section identifies the abstract covariant cost of [Section 10](#) with the moving-carrier production used in the trilogy. It is a concrete phase-I realization of the no-loss theorem, rather than an additional step in the structural proof. No numerical estimate for the gap is needed: compactness upgrades the qualitative equality theorem to a uniform statement.

Lemma 12.1 (Compactness–rigidity gap principle). *Let $\mathcal{K}$ be a sequentially compact space, let $\mathcal{B} \subset \mathcal{K}$ be sequentially closed, and let $\mathfrak{d} : \mathcal{K} \rightarrow [0, \infty]$ be sequentially lower semicontinuous. If*

$$(12.1) \quad \{\mathfrak{d} = 0\} \cap \mathcal{B} = \emptyset,$$

then

$$(12.2) \quad \inf_{x \in \mathcal{B}} \mathfrak{d}(x) > 0.$$

Proof. If the infimum were zero, there would be $x_j \in \mathcal{B}$ with $\mathfrak{d}(x_j) \rightarrow 0$. Sequential compactness gives $x_j \rightarrow x_\infty$ after extraction. Closedness gives $x_\infty \in \mathcal{B}$, while lower semicontinuity gives $\mathfrak{d}(x_\infty) = 0$, contradicting (12.1). $\square$

For the Navier–Stokes application, $\mathcal{K}$ is the sequential closure of the selected windows from one canonical orbit in the full topology of clause (iii) of [Bridge hypothesis 9.2](#); $\mathcal{B}$ is the subset satisfying (9.6); and

$$(12.3) \quad \mathfrak{d}_\nu(\mathfrak{w}) := \int_0^1 \mathfrak{D}_{0, \zeta}(V(\tau)) \, d\tau.$$

Strong local L^3 convergence makes $\mathcal{B}$ closed, while (A.8) makes $\mathfrak{d}_\nu$ lower semicontinuous. Both statements are proved in Proposition A.2. The first assertion of Theorem 11.9 supplies (12.1). Thus the compactness gap uses no semicontinuity assertion about the carrier entropy, the Gaussian Hessian, or the nonlocal ray distance.

Theorem 12.2 (Production gap on a canonical orbit). *Fix the canonical orbit generated by one putative terminal loss. Under alternative (C2) of Bridge hypothesis 11.5, there exists $\delta_* = \delta_*(\text{orbit}) > 0$ such that every selected bad window contained in the terminal tail satisfies*

$$(12.4) \quad \mathfrak{D}_{\text{NS}}(I_j) \geq \delta_*.$$

Proof. Apply Lemma 12.1 to the closure of the selected windows with $\mathfrak{d} = \mathfrak{d}_\nu$, as described above. It gives a number $\delta_\nu > 0$, depending on this orbit, such that $\mathfrak{d}_\nu(\mathfrak{w}_j) \geq \delta_\nu$ for every selected window. Because $\Phi = 1$ on $\text{supp } \zeta$, the carrier cost gives

$$\mathfrak{D}_{\text{NS}}(I_j) \geq \frac{\nu}{2} \int_{I_j} \int \zeta^2 |\nabla V|^2 = \frac{1}{2} \mathfrak{d}_\nu(\mathfrak{w}_j) \geq \frac{1}{2} \delta_\nu.$$

Set $\delta_* := \delta_\nu/2$. □

Corollary 12.3 (One-window entropy estimate). *Every selected bad window $I_j = [\tau_j, \tau_j + 1]$ contained in the terminal tail of the fixed canonical orbit satisfies*

$$(12.5) \quad \mu_{\text{NS}}(\tau_j + 1) \leq \mu_{\text{NS}}(\tau_j) - \delta_*.$$

Proof. Apply (11.25) on I_j and then use (12.4). □

13. COVARIANT EXCLUSION OF SINGULARITIES

Proof of Theorem 1.1. Fix a global finite-energy suitable weak solution (u, p) , and assume that $x_* \in \Sigma_u(T)$. The canonical critical-orbit bridge gives one forward orbit and normalized bad windows with pairwise disjoint interiors $I_j = [\tau_j, \tau_j + 1]$, $j \geq 1$. Relabel them so that $\tau_j + 1 \leq \tau_{j+1}$.

Apply the covariant no-loss cascade bridge. Alternative (N1) is incompatible with retention of the singular target. In alternative (N2), discard the finite prefix before τ_0 . The no-loss window gap gives

$$\mathfrak{m}_{\text{mathrmcov}}(I_j) \geq \delta_{\text{cov}} > 0$$

for every remaining selected window. Countable additivity and disjointness of the windows imply

$$(13.1) \quad \mathfrak{m}_{\text{mathrmcov}}([\tau_0, \infty)) \geq \sum_{j: \tau_j \geq \tau_0} \mathfrak{m}_{\text{mathrmcov}}(I_j) \geq \sum_{j: \tau_j \geq \tau_0} \delta_{\text{cov}} = \infty,$$

contrary to (10.6). Hence $\Sigma_u(T) = \emptyset$. No density of bad times is required; only the existence of infinitely many disjoint target-retaining windows matters.

For $u_0 \in H_\sigma^m(\mathbb{R}^3)$, $m > 5/2$, let T_* be the maximal classical existence time and suppose $T_* < \infty$. Choose a global finite-energy suitable Leray solution. It agrees with the classical solution on $[0, T_*)$ by weak-strong uniqueness and supplies a suitable continuation through T_* . Apply case (b) of Bridge hypothesis 9.2. It produces a canonical orbit even if the concentrating centers escape to spatial infinity. The two no-loss alternatives give the same contradiction as above. Hence $T_* = \infty$. Standard parabolic regularity then gives smoothness on every compact positive time interval, and weak-strong uniqueness identifies the Leray solution with this global classical solution. □

The proof uses each bridge once. The critical-orbit bridge turns terminal loss into infinitely many compact quotient states. The no-loss bridge says that a target-retaining state is either materially reduced or charged to one finite measure. Compactness turns equality rigidity into a positive charge, and countable additivity gives the contradiction. The adjoint entropy, pressure

atlas, moving radius, and Type I/Type II routes are constructions of the two bridges; they do not appear in the final deduction.

Remark 13.1 (Location of the independent problem). The phase-I discharge of [Bridge hypothesis 10.3](#) uses the trilogy’s branch routing. The independent proof has two equation-specific tasks: construct the signed scale-space expenditure without taking absolute values of internal fluxes, and classify its zero-cost material geometries. The first is a conservation problem; the second is a rigidity problem. Neither task is replaced by the formal singular-value observation [\(4.9\)](#).

APPENDIX A. WEAK COVARIANT CALCULUS AND COMPACT STABILITY

This appendix records the standard weak covariant identities and proves the local stability statements used in the gap theorem. It does not supply the strong orbit compactness or the moving-carrier schedule; those remain in the marked bridge contracts.

A.1. Regularization of the equation. Let $S_h v(t) = h^{-1} \int_t^{t+h} v(s) ds$ be a Steklov average on an interior interval, and let v_ε denote spatial mollification. After a diagonal choice,

$$(A.1) \quad u_{\varepsilon,h} \rightarrow u \quad \text{strongly in } L_{\text{loc}}^2 H_{\text{loc}}^1,$$

with weak-* convergence in $L^\infty L^2$. The nonlinear commutator is

$$(A.2) \quad \mathcal{C}_\varepsilon := \operatorname{div}((u \otimes u)_\varepsilon - u_\varepsilon \otimes u_\varepsilon).$$

Since $u \in L_{\text{loc}}^{8/3} L_{\text{loc}}^4$,

$$(A.3) \quad \mathcal{C}_\varepsilon \rightarrow 0 \quad \text{in } L_{\text{loc}}^{4/3} H_{\text{loc}}^{-1}.$$

On a compact repaired interval, spatial mollification commutes with the translation derivative, while

$$(A.4) \quad a[y \cdot \nabla, \varphi_\varepsilon *]V \rightarrow 0 \quad \text{in } L_{\text{loc}}^1 L_{\text{loc}}^2.$$

The spatial commutator is uniformly L^2 -bounded and converges strongly to zero; dominated convergence uses $a \in L_{\text{loc}}^1$ and the local $L_\tau^\infty L_y^2$ bound. Moreover,

$$\|(b_1 - b_2) \cdot \nabla V\|_{H^{-1}(B_R)} \leq c_R |b_1 - b_2| \|V\|_{L^2(B_R)},$$

with the analogous estimate for $(a_1 - a_2)(V + y \cdot \nabla V)$.

Proposition A.1 (Weak validity of the covariant identities). *For every suitable weak solution:*

- (i) $R(u) = 0$ in distributions, and $\mathfrak{D}_\tau^{X,\Lambda} V + \nu A_S V = 0$ on every repaired compact cylinder;
- (ii) the global energy ledger is (3.11);
- (iii) the localized identity is (8.18) for every admissible smooth compactly supported weight;
- (iv) the action-aiming inequalities hold for almost every weak time.

Proof. For item (i), write the mollified equation with residual $\mathcal{C}_\varepsilon$. The linear and pressure terms pass by distributional convergence, the nonlinear residual vanishes by (A.3), and the modulation terms pass by (A.4) and L^1 -approximation of a, b . Item (ii) is the Leray–Hopf energy inequality in the measure form proved in Theorem 3.3. For item (iii), test the suitable local energy inequality by w_Γ , and test the weak momentum equation componentwise by the same scalar weight. The latter identity gives the distributional derivative of $M_\Gamma^{-1} |\int w_\Gamma u|^2 / 2$; subtracting it from the weighted kinetic-energy relation gives exactly (8.18), including the same nonnegative local defect measure. Approximation of temporal indicators gives the asserted one-sided representatives. Item (iv) is Cauchy–Schwarz in the solenoidal Gelfand triple. $\square$

A.2. Compactness of the variational data. Suppose, on compact cylinders,

$$(A.5) \quad V_j \rightarrow V \quad \text{strongly in } L_{\text{loc}}^2 \text{ and } L_{\text{loc}}^3, \quad \nabla V_j \rightharpoonup \nabla V \quad \text{in } L_{\text{loc}}^2,$$

with the corresponding pressure and modulation convergence. For the additional adjoint-weighted conclusions, assume

$$(A.6) \quad m_j \rightarrow m \quad \text{strongly in } L^\infty(B_R \times I), \quad m_j \geq m_* > 0 \quad \text{on } \operatorname{supp} \zeta \times I,$$

and, for normalized adjoint densities,

$$(A.7) \quad \inf_j \inf_\tau Z_j(\tau) > 0, \quad \sup_j \|\tilde{\rho}_j\|_{L^\infty L^2 \cap L^2 H_0^1} < \infty, \quad \sup_j \int_I \int_{B_R} \chi_j^2 |\nabla \sqrt{\rho_j}|^2 dy d\tau < \infty.$$

Proposition A.2 (Stability of the local variational data). *Under (A.5):*

- (i) *the fixed-cutoff means $\bar{V}_{j,\zeta}$ converge strongly in $L^3(I)$, and $C_{\Gamma_j}^\circ(V_j; I) \rightarrow C_\Gamma^\circ(V; I)$;*
- (ii) *the adjoint-weighted kinetic variances converge in $L^1(I)$ whenever $m_j \rightarrow m$ strongly on the full support of the probe;*
- (iii) *the fixed inner Dirichlet floor is lower semicontinuous:*

$$(A.8) \quad \int_I \mathfrak{D}_{0,\zeta}(V) \, d\tau \leq \liminf_{j \rightarrow \infty} \int_I \mathfrak{D}_{0,\zeta}(V_j) \, d\tau;$$

if in addition (A.6)–(A.7) hold, then the adjoint-weighted viscous production is also lower semicontinuous:

$$(A.9) \quad \int_I \mathfrak{D}_{\nu,\Gamma}(V) \, d\tau \leq \liminf_{j \rightarrow \infty} \int_I \mathfrak{D}_{\nu,\Gamma_j}(V_j) \, d\tau;$$

- (iv) *under (A.6), the positivity bound for m on $\text{supp } \zeta$ passes to the limit.*

No lower-semicontinuity claim for the Gaussian Hessian square, the varying-ray distance, or the corrected production is made here, and none is used in the gap theorem. Retaining such terms in a stronger weak-production statement would require identification of the adjoint logarithmic Hessian and all associated localization terms; that refinement is not part of the carrier argument.

Proof. The first assertion follows from strong L^3 convergence and Hölder’s inequality on the fixed support of ζ . The same argument, together with strong convergence of the weights, gives the second assertion. For the third, $\zeta \nabla V_j \rightharpoonup \zeta \nabla V$ weakly in L^2 , which proves (A.8). Under the additional weight convergence, $m_j^{1/2} \nabla V_j \rightharpoonup m^{1/2} \nabla V$ weakly in L^2 ; weak lower semicontinuity of the L^2 -norm gives (A.9). Uniform convergence of the weights gives the final assertion. $\square$

APPENDIX B. WEAK MATERIAL PULLBACK: PROOF CONTRACT

The identities in Proposition 4.1 are classical for a smooth flow. Their use on a compact critical orbit requires a separate weak construction; it is not supplied by the Eulerian compactness statement (9.7).

Contract M.1: regularized material flows. For the smooth approximants used in Appendix A, construct the volume-preserving flows X_ε , deformation tensors F_ε , and metrics g_ε . Prove (4.3)–(4.7) before taking any limit. The approximation must preserve incompressibility and must record the commutator produced by mollifying convection.

Contract M.2: material compactness modulo relabeling. Specify a material relabeling gauge and a topology in which the metric, inverse metric, pulled-back velocity, and pulled-back vorticity have compact subsequences on retained cores. Establish enough convergence to pass the signed weak form of (4.6). No pointwise convergence of F_ε and no classical flow for an arbitrary suitable solution may be assumed. If only a regular Lagrangian flow is available, the proof must state the compressibility, composition, and chain-rule properties used in every pullback.

Contract M.3: compatibility with repaired charts. Prove that material relabeling commutes with the scale–translation cocycle (9.9). Identify the transformation of F , g , and g^{-1} under a parabolic chart change, and show that the zero-expenditure equations are independent of the chosen representative. This contract is required before material reduction can be invoked on a compact normalized window.

APPENDIX C. CANONICAL SINGULAR-GERM EXTRACTION: REPLACEMENT CONTRACT

This appendix will replace the phase-I source attached to [Bridge hypothesis 9.2](#). The replacement is complete only when the following contracts are proved without invoking a final regularity or Type I/Type II exclusion theorem from the trilogy.

Contract B.1: velocity concentration. Starting from the suitable local energy inequality and the velocity–pressure ε -regularity theorem, prove [Theorem 2.1](#). The proof must normalize the pressure by a spatial mean and show that failure of velocity concentration forces both the velocity and reconstructed pressure below the regularity threshold. It must also prove the terminal-loss alternative in case (b) of [Bridge hypothesis 9.2](#): failure of classical continuation yields centers x_j , possibly with $|x_j| \rightarrow \infty$, and scales $r_j \downarrow 0$ carrying a uniform concentration threshold. Finally, it must show that the permitted Galilean reduction leaves the fixed-cutoff oscillation floor (9.6) positive.

Contract B.2: compact descendant selection. Run the ordered concentration–compactness alternatives for singular rescalings and select a branch carrying a fixed threshold η_0 . The proof must terminate at the compact recurrent output required by [Bridge hypothesis 9.2](#); it may not invoke an abstract minimal element without proving existence. Establish inside the selection that:

- (i) two separated retained cores produce the next retained rebase;
- (ii) a same-point cascade produces the next smaller active rebase;
- (iii) diffuse or exterior escape contradicts the retained threshold after pressure reconstruction;
- (iv) a pressure-only carrier either produces a velocity descendant or lies below the local regularity threshold;
- (v) constant velocity modes disappear from $\mathcal{C}^\circ$, while affine pressure modes are recorded in the pressure atlas.

Contract B.3: coherent forward orbit. Construct one absolutely continuous chart on $[0, \infty)$, rather than independent charts on selected windows. Prove the repaired equation, $\Lambda(\tau_j) \rightarrow 0$, the cocycle law, and the existence of infinitely many unit windows with pairwise disjoint interiors satisfying (9.6). The construction must show that every reselection represents the same physical solution and preserves the quotient state. It must work both for bounded centers and for the escape-to-infinity alternative in case (b).

Contract B.4: compactness and persistence. Derive (9.5), strong local L^3 compactness, pressure compactness modulo (9.4), and passage of the fixed-cutoff $\mathcal{C}^\circ$ to the limit. Prove sequential compactness of the entire closure of the selected windows, not only subsequential compactness of the original sequence. Together with Contract C.3, obtain strong convergence and positivity of the fixed-radius local weights in (A.6) when the optional adjoint diagnostic is retained. The orbit conclusions must hold with constants depending only on the normalized class $(\eta_0, \mathbf{M}, \nu)$.

PHASE-I SOURCE. The precise precursor results are listed in the trilogy source ledger in the introduction.

INDEPENDENT PROOF CONTRACT. Contracts B.1–B.4 specify the synthesis needed for the new single-orbit statement, including the additional oscillation, coherence, and escape-to-infinity clauses.

APPENDIX D. GAUGE, PRESSURE, AND ADJOINT CONTROL: REPLACEMENT CONTRACT

Contract C.1: canonical scale–center gauge. Use the four-component map (6.8) and Jacobian (6.9) to solve for scale and center on every nondegenerate retained window. Prove

uniqueness on the local slice, absolute continuity of X, Λ , and $a, b \in L^1_{\text{loc}}$. At degeneracy, the ordered selection in Contract B.2 must either supply a new nondegenerate slice or leave the retained class. No moving condition $\bar{V}_m = 0$ is to be constructed; Galilean invariance is already built into (6.8), (8.4), and (9.2). When differentiating the gauge, include the derivatives of both fixed-cutoff means in (6.7).

Contract C.2: pressure atlas. For nested inner balls prove

$$P = P_{\text{loc}} + H, \quad -\Delta P_{\text{loc}} = \partial_i \partial_j (\zeta V_i V_j), \quad \Delta H = 0,$$

with compatible spatial means. Establish uniform Calderón–Zygmund bounds for P_{loc} , harmonic interior estimates for H , stability under chart transitions, and convergence in $L^{3/2}_{\text{loc}}$.

Contract C.3: normalized adjoint family. For the optional Perelman diagnostic, strengthen Lemma 7.2 by proving a uniform lower bound for Z , the Fisher estimate in (A.7), and compactness of the normalized density on padded windows. Use the covariant scale (7.7) and verify (7.6). For every entropy cutoff χ , choose a nested $\hat{\chi}$ equal to one near $\mathcal{A}_\chi$ and prove the Fisher bound required by (11.17). These estimates refine the geometric interpretation but are not inputs to the moving-carrier contradiction.

Contract C.4: summable transport errors. Construct a nondecreasing radius $R(\tau) \rightarrow \infty$ on the coherent orbit. Derive (11.22) directly from the suitable local energy inequality and prove that the diffusion-shell, convection, pressure, and translation components are summable on this schedule. Pressure must be handled by the local/harmonic atlas of Contract C.2, uniformly for every finite-valued moving radius above the scheduled lower bound. No adjoint normalization or terminal-data reset occurs in this contract.

PHASE-I SOURCE. The phase-I local bounds and repaired-gauge estimates are drawn from the pressure atlas and corrected moving-cutoff results in [4].

INDEPENDENT PROOF CONTRACT. Contracts C.1–C.4 specify the estimates that must be derived to support the covariant diagnostic and the moving carrier without depending on that paper.

APPENDIX E. CANONICAL MOVING-CARRIER ALTERNATIVE: PROOF CONTRACT

The monotonicity calculation is already Theorem 11.4. This appendix must prove the global alternative in Bridge hypothesis 11.5. The phase-I source is the trilogy’s canonical pressure decomposition and modulation system, retained pressure-tail and tailored-radius lemmas, moving-family control, transition and scale routing, corrected monotonicity, cost-divergence exclusion, and thin negative-drift closure. Their citations in the introduction are placeholders for the independent derivation below.

Contract D.1: exact carrier ledger. Starting from the suitable local energy inequality in the repaired equation, derive every term of (11.22), including the $\partial_\tau \Phi$ term and the signs of the two scale contributions. Prove the gauge invariance of the pressure flux and the favorable sign of the suitable energy-defect measure. This recovers Theorem 11.4 without appealing to a companion monotonicity theorem.

Contract D.2: pressure-tail and radius scheduling. On successive unit windows, use critical tightness to choose radii $R_n \rightarrow \infty$ with a prescribed summable exterior L^3 tail. Prove uniform vanishing of the pressure flux for every moving radius above R_n , using the Calderón–Zygmund local part and harmonic pressure atlas separately. Interpolate the R_n monotonically and show that the diffusion-shell, convection, pressure, and translation summands of B_{mov} lie

in L^1 . A failure must be assigned to a specific pressure, tightness, exterior, or modulation row before the retained carrier branch is declared.

Contract D.3: transition and positive-scale routing. For the canonical interpolated radius, analyze the transition summand $|\int |V|^2 \partial_\tau \Phi|$ and the positive scale-shell summand $a_+ \int |V|^2 (-y \cdot \nabla \Phi)$. Prove that each is either summable on a later tail or produces a normalized occupation sequence in one of the already named exterior, recentring, positive-scale, compact-cost, or scale-collapse rows. The alternatives must be exhaustive and stable under the compact topology of (A.5); mere failure of integrability may not be replaced by an unjustified unit-window lower bound.

Contract D.4: negative scale drift and final synthesis. Treat $a_- \int |V|^2 \Phi$ by the finite-cost/divergent-cost dichotomy. On diffuse normalized drift blocks, prove that $a_-(V + y \cdot \nabla V) \rightarrow 0$ in $L^1 H_{\text{loc}}^{-1}$ on selected unit windows and that its compact local-energy pairings vanish. Route persistent blocks to compact scale-collapse or cost divergence, and route all first failures to their named local rows. After these routes, either a named branch is already closed, giving alternative (C1), or all seven summands of (11.22) are integrable on one tail, giving alternative (C2). This proves the bridge without constructing an adjoint reset cocycle.

APPENDIX F. SIGNED SCALE-SPACE CURRENT: INDEPENDENT PROOF CONTRACT

The moving carrier estimates every boundary term in absolute value. This is stable at suitable-weak regularity, but it destroys cancellation when the outgoing flux of one cell is the incoming flux of a smaller cell. The independent covariant proof must retain that cancellation.

Contract S.1: compatible scale-space cells. Construct a locally finite rooted family of nested parabolic cells $\{Q_\alpha\}_{\alpha \in \mathcal{T}}$ along the coherent critical orbit, together with smooth kinetic-energy tests whose spatial and temporal boundaries form a compatible partition. Every retained window must contain a distinguished cell, and every infinite target-approaching sequence must determine an infinite path in $\mathcal{T}$. Sibling cells may overlap only through a prescribed bounded-multiplicity buffer. The construction must be invariant under the scale-translation cocycle and the fixed Galilean centering.

Contract S.2: signed edge-current identity. For each finite subtree $\mathcal{T}_0$, sum the suitable local energy identities before taking positive parts. Write the result in the form

$$(F.1) \quad K_{\text{leaf}}(\mathcal{T}_0) + \mathcal{D}(\mathcal{T}_0) + \mathcal{M}(\mathcal{T}_0) \leq K_{\text{root}} + \mathcal{I}_{\text{ext}}(\mathcal{T}_0),$$

where $\mathcal{D}$ is viscous expenditure, $\mathcal{M}$ is the favorable suitable-energy defect, and $\mathcal{I}_{\text{ext}}$ is input through the outer boundary. Prove exact cancellation of every internal convection and pressure current between a parent and its children. The cancellation must include the harmonic pressure representative and must be unchanged by adding a function of time to the pressure.

The derivatives of scale, center, and material relabeling are to appear as edge coboundaries. Their sum over $\mathcal{T}_0$ must reduce to root and leaf terms. Estimating these terms separately on each cell does not satisfy this contract.

Contract S.3: finite covariant expenditure. Pass monotonically through finite subtrees in (F.1). Prove that the root energy and true exterior input are finite and define a nonnegative Radon measure $\mathfrak{m}_{\text{mathrm{cov}}}$ with (10.6). Establish (10.7) for a lower-semicontinuous window functional whose zero set is precisely the lossless material system. A normalized quantity obtained by dividing the physical dissipation by a vanishing scale does not constitute a finite measure; the scale-critical normalization must come from the tree telescoping itself.

Contract S.4: contact and exhaustive descent. Prove that retained oscillatory contact either reaches a materially reduced cell or continues to a child carrying the same fixed target threshold. Rule out infinite hopping among siblings by bounded overlap and target additivity. This produces the infinite path required in Contract S.1 without the local Type I/Type II branch list. Together, Contracts S.1–S.4 construct the measure and descent part of [Bridge hypothesis 10.3](#).

APPENDIX G. LOSSLESS MATERIAL GEOMETRY: INDEPENDENT RIGIDITY CONTRACT

This section records the central equation-specific theorem still required by the independent proof. It is the structural replacement for the trilogy’s terminal branch closures.

Contract R.1: compact lossless limit. Let $\mathbf{w}_j$ be normalized target-retaining windows with $\mathfrak{d}_{\text{cov}}(\mathbf{w}_j) \rightarrow 0$. Using Appendices [A](#) and [B](#), extract one limit $(V_\infty, P_\infty, F_\infty, \mathbf{g}_\infty, \boldsymbol{\Omega}_\infty)$ modulo the declared spatial, scale, Galilean, pressure, and material-relabeling gauges. Derive every zero-expenditure equation from lower semicontinuity; no strong convergence of the deformation tensor may be inserted as an additional hypothesis.

Contract R.2: bounded-deformation rigidity. If $\mathbf{g}_\infty$ and $\mathbf{g}_\infty^{-1}$ are locally bounded on the retained core, use uniform parabolicity of the pulled-back diffusion and the lossless equations to prove that V_∞ is spatially constant there. Conclude that its fixed-cutoff oscillatory target vanishes.

Contract R.3: anisotropic reduction. If the condition number of $\mathbf{g}_j$ diverges, use [\(4.9\)](#) to identify the enhanced compressed directions. Prove the following exhaustive alternative:

- (i) the pulled-back velocity or vorticity has nonzero variation in an enhanced direction, which gives positive covariant expenditure; or
- (ii) those transverse derivatives vanish, the associated limiting distribution is invariant and integrable, and the induced equation is a two-dimensional, shear, or other explicitly classified stretching-depleted system.

For every system in item (ii), prove regularity on the retained cylinder and a continuation or descent statement that closes the terminal singular branch. A regular reduced state may have nonzero oscillation on a fixed unit window, so regularity may not be replaced by a claim that $\mathcal{C}^\circ$ vanishes. Alignment of vorticity with one strain eigenvector does not by itself prove integrability and may not be used as a substitute for this classification.

Contract R.4: equality closure. Show that the bounded and anisotropic alternatives are closed under the window topology and exhaust every zero-cost sequence, including eigenvalue crossings and material-frame rotation. Every zero-cost limit must either be oscillation-removable, giving [\(10.8\)](#), or trigger the certified material-reduction closure in alternative (N1). The adjoint strain–Gaussian square may be used to identify a whole-space equality model, but a localized adjoint density may not be declared Gaussian. Contracts R.1–R.4, combined with the signed current contracts, complete the independent proof of [Bridge hypothesis 10.3](#).

APPENDIX H. LOCALIZED EQUALITY AND THE ROLE OF THE GAUSSIAN

For the phase-I moving-carrier realization, the proof of [Theorem 11.9](#) uses only the coercive fixed-cutoff Dirichlet floor and connectedness of $\{\zeta > 0\}$. These properties are closed under the convergence already required by the orbit bridge. The independent signed-current realization has a larger apparent equality class; its reduction to target-null states is the content of [Appendix G](#).

The Gaussian statement has a separate and more limited role. For a smooth positive whole-space probability density, vanishing of $\mathfrak{D}_G$ gives $\nabla^2 f = (2\nu\theta)^{-1}I$, and normalization and

integrability then give the heat kernel (7.18) up to translation. A localized Dirichlet adjoint density is not that whole-space Gaussian. The manuscript therefore uses Gaussian rigidity only as the equality model for the formal adjoint entropy, not as a premise in the local oscillation contradiction.

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